

A Mini Project with Seminar

On

**UPI Transaction based Fraud Detection System using Generative
AI and Machine Learning Techniques**

Submitted in partial fulfilment of the requirements for the award of the

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in

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by

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CERTIFICATE

This is to certify that the project entitled “**UPI Transaction based Fraud Detection System using Generative AI and Machine Learning Techniques**” is submitted by **Md. Shoaib Nadeem (23241A67A0), Ch. Vishal Abhishek (23241A6774) and L. Shiva kumar (23241A6797)** in partial fulfillment of the award of degree in BACHELOR OF TECHNOLOGY in Computer Science Data Science during the academic year 2025-2026.

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DECLARATION

We hereby declare that the mini project titled “**UPI Transaction based Fraud Detection System using Generative AI and Machine Learning Techniques**” is the work done during the period from **12th December 2025 to 23rd April 2026** and is submitted in the partial fulfillment of the requirements for the award of degree of Bachelor of Technology in Computer Science and Engineering (Data science) Gokaraju Rangaraju Institute of Engineering and Technology (Autonomous under Jawaharlal Nehru Technology University, Hyderabad). The results embodied in this project have not been submitted to any other University or Institution for the award of any degree or diploma.

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ABSTRACT

Growing proliferation of digital payments particularly unsupervised UPI operations in recent years has lead to increased instances of financial frauds worldwide and emphasizes need for a reliable fraud detection framework. Traditional supervised machine learning methods are often limited by inadequacy of fraudulent samples in the dataset and excessive skewness in class distributions. Therefore, this project introduces a fraud detection model with Generative Artificial Intelligence empowered advanced machine learning methods. A Conditional Generative Adversarial Network (CGAN) is used to produce authentic fake fraudulent samples based on transaction behaviours in similar context, facilitates enhancement of training data in a balanced manner, trainest robust classification model using XGBoost to effectively identify fraudulent candidates while NGBoost estimates fraud probability and prediction uncertainties. Also, model transparency is ensured by employingshining-ing (SHAP) views. The system is tested on IEEE-CIS Fraud Detection Dataset of 20000 transactions, with roughly 10% of frauds and results suggest that the model performs markedly better than typical models, which do not seem to incorporate fake generated data information. It attains high fraud detection rate of 84.4% with factually correctness ratio of 90.6% and an F1 measure of 87.4%, a strong indicator of its competence in fraud detection. To summarize, the model boost detection robustness and false negatives, and yields interpretable, principled andCratable predictions. These inform features make it applicable in real-time digital payment scenarios such as UPI and other online financial transactions.

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LIST OF ACRONYMS

S.NO	Acronym	Abbreviation
1	AI	Artificial Intelligence
2	API	Application Programming Interface
3	CGAN	Conditional Generative Adversarial Network
4	CSV	Comma Separated Values
5	GAN	Generative Adversarial Network
6	ML	Machine Learning
7	NGBoost	Natural Gradient Boosting
8	SHAP	SHapley Additive Explanations
9	UPI	Unified Payments Interface
10	XAI	Explainable Artificial Intelligence
11	XGBoost	Extreme Gradient Boosting

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CHAPTER 1

INTRODUCTION

1.1 Introduction to the Project Work

In today's world the way people make payments has changed a lot with the use of real-time digital payment systems like Unified Payments Interface. Unified Payments Interface has become one of the popular ways to make payments because it is fast easy to use and accessible to everyone. However with more people using payments there has been a big increase in fake transactions, which is a big problem for banks and users. Old systems that detect fraud mostly use rules and basic machine learning. These methods work to some extent. They often miss complex fraud patterns that are always changing. One big problem with detecting fraud is that there are few fake transactions in the data so the machine learning models get better at finding real transactions but not good at finding fake ones. This means they are not very good at detecting fraud especially when it comes to finding all the transactions.

To solve these problems new methods like Generative Adversarial Networks have been developed. Conditional Generative Adversarial Networks are especially useful because they can create examples of fraud based on what we know about transactions. This helps make sure the system is good at finding transactions. Also new systems need to be transparent and explainable so people can trust them. This project is about making an Unified Payments Interface fraud detection system that uses Generative AI and machine learning. The system uses Conditional Generative Adversarial Networks to balance the data, XGBoost to classify transactions, SHAP to estimate the risk of fraud and SHAP to explain how the system works. By combining these methods the system is good at finding transactions, accurate and easy to understand. The system provides a solution for detecting fake transactions in digital payment systems, like Unified Payments Interface.

1.2 Problem Statement

The way people make payments has changed rapidly in recent years, especially with the growth of Unified Payments Interface (UPI). While this has made transactions faster and more convenient, it has also led to a noticeable rise in fraudulent activities that are becoming harder to detect. Most existing fraud detection systems rely on traditional machine learning models or rule-based methods, and these approaches face several challenges. One of the biggest issues is that fraudulent transactions are very few compared to genuine ones. Because of this imbalance,

models tend to perform well on normal transactions but struggle to correctly identify fraud. Another concern is the heavy focus on accuracy as a performance measure. A model can appear highly accurate overall, yet still miss many fraudulent cases, which can result in serious financial losses.

There are also limitations related to transparency and decision-making. Many current systems behave like black boxes, making it difficult to understand how a prediction is made, which reduces trust in their outputs. In addition, these systems often do not provide information about how likely a transaction is to be fraudulent, even though such insights are important for taking action. Fraudsters are constantly changing their strategies, which means detection systems must also be flexible and able to adapt over time. For these reasons, there is a need for a more balanced and reliable approach—one that can effectively detect fraud despite limited examples, provide clear explanations for its predictions, and offer a measure of confidence in its results. Such capabilities are essential for modern digital payment environments like UPI.

1.3 Purpose and Objectives of the Proposed Work

The primary objective of this project is to create an effective fraud detection system for United Payments Interface (UPI) transactions by blending the power of machine learning with Generative AI. The system aims to detect fraudulent transactions efficiently while maintaining clear, dependable explanations for its predictions. This is essential, considering fraud detection systems operate within high-stakes financial settings.

To achieve this, the proposed system focuses on several objectives, they are as follows:

- To develop a UPI transaction fraud detection system capable of identifying fraudulent activities.
- To address class imbalance in fraud data for synthetic fraud sample generation.
- To implement Classification algorithm that reduces false negatives and improves fraud detection accuracy.
- To incorporate an algorithm for probabilistic fraud risk estimation, enabling risk-aware and confidence-driven transaction decision-making.
- To provide explainable fraud predictions for transparent and trustworthy decision-making.

1.4 Scope of the Proposed Work

The goal of this project is to build a system that can detect fake transactions on UPI using AI and machine learning. This system will look at transaction data find activities and give predictions that show the level of risk and explain why. The system also uses SHAP to explain how it makes decisions about fraud detection this means it can show which factors are most important in identifying transactions. The project only uses existing data, which does not involve real-time deployment or integration with live banking systems. The data used is from the IEEE-CIS fraud detection dataset. The system aims to improve how well it detects fraud. However, it may not catch all transactions because new types of fraud are being created and the data may not cover everything. The focus is on making the system more accurate recalling fake transactions and being more interpretable. The project works within the limits of the data and models. UPI transactions are analysed to improve detection. Machine learning models are used for fraud detection on UPI transactions. The system provides insights into fraud detection decisions, for UPI transactions.

1.5 Methodology and Approach

The proposed fraud detection system is built with a simple goal in mind: to reliably identify suspicious UPI transactions while making sure the decisions are understandable. It starts with a dataset of about 20,000 transactions, which includes both genuine and fraudulent cases. Since real transactions are far more common than fraudulent ones, the data is first cleaned and prepared carefully. To handle this imbalance, the system uses a Conditional Generative Adversarial Network (CGAN). This technique essentially learns the pattern of fraudulent transactions and generates new, realistic fraud examples, helping create a more balanced dataset for training. Once the data is ready, the system uses XGBoost to learn patterns and classify transactions as either real or fraudulent. To make the predictions more meaningful, NGBoost is added on top of this, which not only predicts whether a transaction might be fraud but also gives a confidence level for that prediction. This makes the system more practical and trustworthy, especially in situations where decisions need to be made carefully. Another important aspect of the system is transparency. Using SHAP, the model can explain why it flagged a transaction as suspicious by highlighting the key factors that influenced its decision. Altogether, the system follows a clear and connected process—from preparing data and generating additional samples to training models, making predictions, and explaining results

resulting in a fraud detection approach that is not only effective but also easy to understand and trust.

1.6 Organization of the Project Report

The project report is organized into six chapters, each focusing on a specific aspect of the proposed UPI fraud detection system. This structure ensures a clear and logical flow, allowing the reader to understand the problem, the approach, and the outcomes in a systematic way.

Chapter 1: Provides an overview of the project. It explains the background of UPI transactions and the growing need for fraud detection, clearly defines the problem being addressed, and outlines the objectives of the system. It also describes the scope, methodology, and overall organization of the report.

Chapter 2: Presents a review of existing research in the field of fraud detection. It discusses various machine learning and generative AI approaches used by other researchers, highlighting their strengths as well as their limitations. This chapter helps in understanding the gap that the proposed system aims to address.

Chapter 3: Explains the design and working of the proposed system in detail. It describes the overall architecture, key components, and the algorithms used, including CGAN for data generation, XGBoost for classification, NGBoost for probabilistic prediction, and SHAP for explainability. It also includes UML diagrams and system requirements necessary for implementation.

Chapter 4: Focuses on the experimental evaluation of the system. It describes the dataset, experimental setup, and performance metrics, followed by a detailed analysis of results using graphs, tables, and comparisons. This chapter demonstrates how effectively the system detects fraudulent transactions.

Chapter 5: Summarizes the work carried out, key findings, and overall performance of the system. It also suggests possible improvements and future directions to enhance the system's accuracy, scalability, and applicability.

Chapter 6: Contains supplementary material such as sample code, additional data, and supporting information that provides further clarity and completeness to the project.

CHAPTER 2

LITERATURE SURVEY

2.1 Summary of existing approaches

With the rapid development of electronic payment and financial technology, fraud detection is always one of the research hotspots in recent years. Imbalanced dataset of real and fraud transactions may affect the precision of a model, which makes others focus on the generation of fraud samples. Some studies are introduced to generate better fraud samples by using generative models. For example, an improved VAE-GAN model [1] addresses the issue of generating more realistic samples by merging Variational Autoencoders with Generative Adversarial Networks, which benefits from the better feature learning for generating synthetic fraud samples. Similar to these mechanisms, GAN-based data augmentation [2] has been proposed and proven to be able to generate needs of fraud samples and accelerate the training of classifiers with improved accuracy.

Conditional GANs (CGAN) take this a step further and condition other relevant information onto the fake samples such as transaction amount, transaction time, merchant category [3]. Conditionally using this information in generation helps the fake samples to be more accurate and specific to the scenario, which enhances its effectiveness during training. Not only that, but CGANs are known to enhance sample quality and perform better than a traditional GAN or a SMOTE method. Semi-supervised GAN [4] not only takes into account the labeled data, but also takes into account the unlabelled data. This method is particularly advantageous when the number of fraud labels is small. Furthermore hybrid approaches such as WDAE-GAN [5] combine Denoising Autoencoders with GANs by deepening the data before generation to remove noise, thereby increasing the quality of samples. Attention-based GAN [6] den knock out irrelevant details during image generation by focusing on the most important features for fraud detection and generation of realistic samples. However, such methods are computationally intensive and require hyper-parameter fine-tuning. In addition to generation models, classification models are also important components of a fraud detection system. Among these, XGBoost [7] has been perceived as the most effective algorithm for dealing with fraud transaction data due to its speed, accuracy and scalability. By using a Gradient boosting method that offers iterative improvement, this method is well suited for big transaction datasets in the real world. However, XGBoost does not tackle class imbalance issue, making data balancing techniques even more significant to use in conjunction with the model. Systems such

as Scarff [8] explores the ability of scalable frameworks to use boosting models to achieve high accuracy at scale in fraud detection. NGBoost [9] introduces probabilistic prediction capabilities into the model so that predictions can also be accompanied with uncertainty measurements. Such capabilities are also supported by research [10], which further shows that NGBoost can be used effectively in fraud detection.

Baseline models: Random Forest [16] and Logistic Regression [17] have been traditionally employed as benchmark models in fraud detection. Random Forest shows great performance in fraud detection as it combines several decision trees, it is also robust, while Logistic Regression provides a simple model that can be easily interpreted. Nevertheless, they are not able to learn complex combination of transaction data. Deep neural networks [18] successfully learn on the complex feature interactions when they are trained on data, providing high accuracy in fraud detection task. Nevertheless, deep neural networks require a huge amount of data to train on and they are hard to interpret. Unbalanced data learning is a key problem in fraud detection. SMOTE [11] is a simple idea to generate new samples for unbalanced learning setting, by interpolating known examples from the target class. Nevertheless, since it ignores relations among features in real world, this can generate noisy or overlapping data. In [12], it is highlighted the importance of suitable evaluation metrics for imbalanced data learning other than accuracy such as recall, precision and F1-score. Concept drift in transaction data can be caused by fraudster evolution who are also changing their strategies. In [15], it was considered the problem of detecting and adapting to adversarial drift of transaction data as a new concept of concept drift.

Explainability has been a crucial aspect of present fraud detection systems. Explainable AI techniques [13] try to improve the transparency by explaining the decision making of the models. SHAP (SHapley Additive exPlanations) [14] is one of the most widely used techniques for understanding a prediction. The technique analyses the impact of each feature towards a specific prediction, so it can be used to justify why a transaction was recorded as an illegal activity. While SHAP offers straightforward and consistent explanations, it is computationally demanding when used on big data. Past research have also studied hybrid techniques that enhance the performance of fraud detection by integrating multiple approaches. Hybrid machine learning models [19] are designed to enhance the accuracy by combining different models that share their abilities, but the overall system structure become more complex. Transaction behaviour analysis [20] used to discover abnormalities in transactions logs, thus combining a second layer of fraud detection to the base system. The above survey makes it

clear that none of the approaches can fully resolve the problems faced in fraud detection. Different approaches such as generative models for addressing class imbalance, machine learning algorithms for fraud classification, probabilistic models for estimating prediction uncertainty, and Explainable AI for identifying feature importance in AI decision processes are to be integrated to build a more robust system. Their advantages and limitations can be observed in the proposed system, which implements CGAN to generate data, XGBoost to classify fraud, NGBoost to make probabilistic prediction, and SHAP to interpret data. This approach solves the main challenges faced in UPI fraud detection, such as imbalance, accuracy, and explainability.

Table 1 represents a summary of existing approaches used in fraud detection. It highlights different methodologies such as GAN-based models, machine learning algorithms, and hybrid techniques along with their drawbacks and outcomes

Table 1: Existing Approaches

Ref. No.	Authors	Methodology	Drawbacks	Result
1.	VAE-GAN Model (2023)	Hybrid architecture combining Variational Autoencoder (VAE) for latent feature extraction and GAN for generating realistic synthetic fraud samples to handle class imbalance.	High computational complexity; requires hyperparameter tuning.	Higher Accuracy
2.	GAN-based Fraud Detection (2023)	Trains generator–discriminator network to learn fraud data distribution and augment dataset with synthetic samples for improved classification.	Sensitive to training instability; may produce noisy data.	Data stability

3.	CGAN for Fraud Detection (2023)	Conditional GAN uses transaction attributes (amount, time, location) as input conditions to generate context-aware synthetic fraud samples.	Complex training; risk of mode collapse.	Improved precision
4.	Semi-Supervised GAN (2025)	Combines supervised learning with GAN where discriminator performs classification using both labeled and unlabeled transaction data.	Difficult training balance; higher complexity.	Improved recall
5.	WDAE-GAN (2025)	Integrates denoising autoencoder with GAN to remove noise before generating high-quality fraud samples.	Resource intensive; longer training time.	High performance
6.	Attention-based GAN (2024)	Incorporates attention mechanisms within GAN to focus on important transaction features for better synthetic data generation. Also improves feature weighting by prioritizing critical fraud indicators.	Very high computational cost.	High recall
7.	Chen & Guestrin (2016) – XGBoost	Gradient boosting framework that builds ensemble decision trees sequentially using error minimization and regularization.	Requires external imbalance handling.	High accuracy

8.	Scarff Framework (2023)	Ensemble-based fraud detection framework evaluating multiple boosting models for scalability and robustness.	No data balancing techniques used.	Good performance
9.	NGBoost (2020)	Uses natural gradient boosting to estimate full probability distributions instead of point predictions.	Computationally expensive.	Good calibration
10.	NGBoost Fraud Study (2023)	Applies NGBoost on fraud datasets to generate probabilistic outputs for risk-aware decision-making.	High training cost; tuning required.	Improved probability accuracy
11.	SMOTE (2002)	Generates synthetic minority samples using interpolation between nearest neighbors in feature space. It increases dataset balance without requiring additional real data.	May create overlapping classes; unrealistic data.	Moderate
12.	Imbalanced Data Study (2009)	Analytical study focusing on evaluation metrics like recall, precision, and F1-score for imbalanced classification problems.	No implementation model.	Not applicable
13.	Explainable AI (2025)	Integrates model interpretation techniques to explain predictions and improve transparency in fraud detection.	Adds computational overhead.	Improved trust

14.	SHAP (2024)	Uses Shapley values from game theory to compute feature contribution for each prediction.	Computationally expensive for large datasets.	Accurate explanations
15.	Drift Detection (2022)	Uses ensemble techniques to detect concept drift in fraud patterns and adapt models dynamically.	Does not solve imbalance issue.	Stable
16.	Random Forest	Uses multiple decision trees with bagging to improve prediction accuracy and reduce variance. It aggregates outputs from multiple trees for robust classification.	Overfitting possible; less interpretable.	Good accuracy
17.	Logistic Regression	Uses linear probability model to classify fraud based on weighted feature inputs.	Cannot capture non-linear patterns.	Moderate
18.	Deep Learning Models	Uses multi-layer neural networks to learn complex patterns and feature interactions in large datasets.	Requires large data; black-box model.	High accuracy
19.	Hybrid ML Models	Combines multiple ML algorithms (ensemble/hybrid approach) to improve robustness and detection accuracy.	Complex implementation and maintenance.	Strong

20.	Behaviour Analysis	Detects fraud by analyzing transaction sequences and identifying anomalies in user behaviour patterns.	Needs large historical data.	Good
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2.2 Drawbacks of existing approaches

Despite all the advancements made in the detection of fraud through the years, many methods in practice today still have various obvious shortfalls. The most significant of these is the problem of imbalanced data, as the majority of records are labelled genuine, while only a small minority is fraudulent. Existing machine learning techniques, which explicitly solve the imbalances data problem typically perform poorly at catching instances of fraud. SMOTE attempts to alleviate this by synthesizing new data from each input sample, but the new samples generated can often be inexact which hinders performance; whereas using an existing GAN model produces higher quality data, but is far more complex to implement, tune and is computationally expensive. The use of deep learning models can lead to high accuracy, but they are often “black-boxes”, leading to reduced trust.

Other issues include the fact that many of the existing systems are trained with a focus mainly on accuracy rather than recall. This is widely accepted when the financial loss is relatively small, but it makes the efficiency of recall much more vital in the context of fraud detection, where a few missed fraudulent transactions could equate to very large losses. Though this is problematic in many existing systems, it is often neglected. Furthermore, most of these systems do not output any probabilistic-based results (e.g. Confidence scores), which have been found to be very useful during real business decision making. This makes a user less able to judge how risky a certain transaction really was. With that, being able to explain why the fraud was detected is another major concern, due to the fact that a large amount of existing systems cannot clearly explain why an incident was caught. There are also many application-related problems, for example with the data and system design. Since transaction data streams are prone to contain erroneous information, be missing or contain errors, the data can have a huge impact on the ability of the model performance to generate real fraud alerts.

CHAPTER 3

PROPOSED METHOD

3.1 Architecture Diagram and Methodology

The pipeline for fraud detection as shown in figure 1 below is performed by collecting the transaction data, data pre-processing and use of CGAN module to balance the imbalanced data by generating synthetic class minority data and using XGBoost to do classification, using NGBoost for risk prediction and finally using SHAP to explain the results and output the results with fraud statues and risk score.

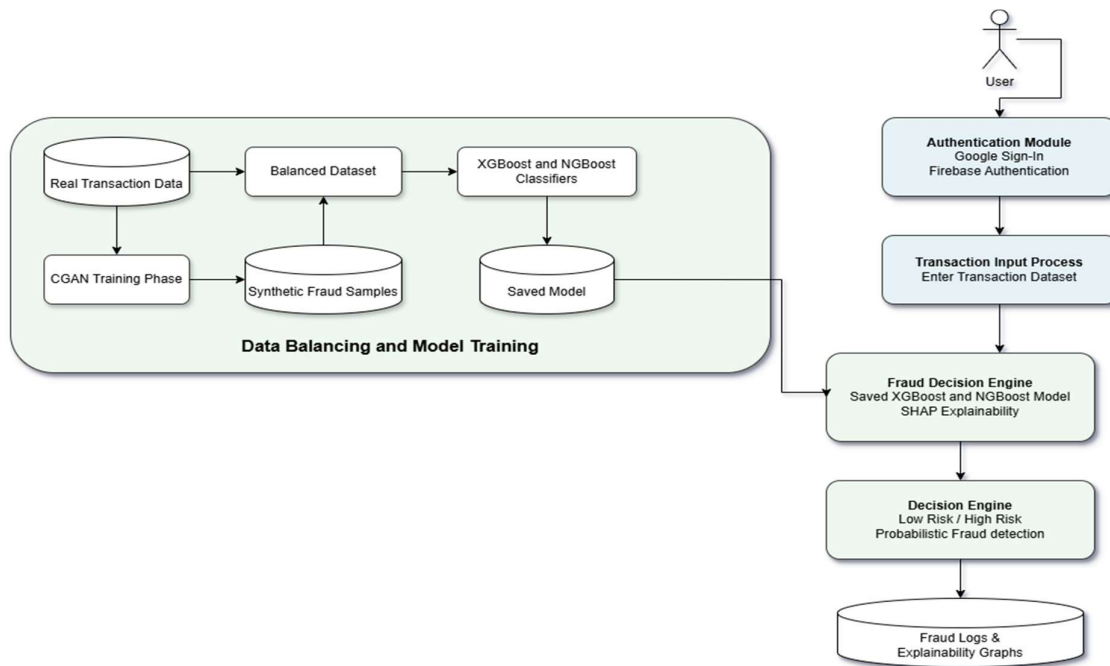


Figure 1: Architecture of proposed Fraud Detection System

The architecture of the fraud detection system based on transaction dataset architecture proposed in this paper consists of a well-defined and sophisticated pipeline that brings together multiple novel and latest technology APIs into a unified and integrated flow. The overarching concept of this UPI transaction fraud detection system architecture is that it aims to be substantially accurate, trustworthy, open, forward looking, and scalable by efficiently combining the authoritatively dynamic UPI transaction data stream analysis, generative AI, specialized ML algorithms, and the transparency enableible explainable AI in a pipelined workflow.

The first step is the collection of data related to transactions, which are the key of the entire system. It takes data from two sources: from the set of stored data, which in the framework of this project was the IEEE-CIS dataset and from a user, who can provide CSV files and how to simulate transactions. Data are constituted from some parameters such as transaction amount and time, transaction merchant and its place of origin. These parameters are important since it is not usual to make a fraud decision based on one parameter, but in the decision are involved relationships between one or more parameters. For example, a transaction is not necessarily suspicious if alone seem regular, but is not when it is considered together with the transaction time and the origin place of it. This is the first point in order to the system get rich and accurate data.

After the data is collected, it is passed to the preprocessing step, which is an essential point of data analysis. Transaction data may contain anomalies, null values, and noise. These factors may have an adverse effect on the performance of the machine learning models. For this purpose, some preprocessings are performed. Null values are estimated using some appropriate techniques (imputation). Some unimportant or redundant features are excluded. Feature extraction is performed, i.e., some new features are estimated based on the existing features. For example, the system might be able to calculate how often certain features are experienced during that period, e.g., the number of transactions on one day, or the way users behave. The system is also able to perform the conversion of numerical data to categorical data and vice versa, as well as standardization or normalization of numerical data, which are needed because of the requirement of the machine learning algorithms. The empty data structure after preprocessing is well-formed and ensures a high level of high quality and it is understandable for the computer. This step is necessary for all the subsequent processes. The most important factor that makes fraud detection challenging is the already mentioned unbalanced data. Normally the number of fraud cases in a real dataset is much lower than those of normal transactions. This causes a bias of machine learning algorithms to make a decision that all cases are normal. As a result, the ability to identify fraud decreases. That is why the architecture contains a specific block called Conditional GAN which is able to generate fraud samples. It is a neural network structure, which is formed by two neural networks discriminator and generator working in a competition. The generator uses random numbers and some conditional attributes (e.g., time or amount of transaction) to create artificial data, which is then tried to be classified by the discriminator as real or fabricated. Both networks learn in a competition, i.e., the generator learns how to generate more realistic data, and the discriminator enhances at

classifying real or fake data. After some iteration this kind of network yields very good quality synthesized data, which can be used together with the initial data as a clear and well-structured dataset, which is a great input for machine learning algorithms.

Once the data has been balanced, the cleaned data flow into the machine learning layer, which is the core decision maker of this system. In this layer, two very powerful algorithms will be used; XGBoost and NGBoost. The XGBoost engine will determine if a transaction is fraud or not; it is a Gradient boosting algorithm which produce through multiple decision trees, sequentially, a final classification/regression decision, learning from the errors of the previous trees. As a result, it allows learning complex patterns in structured data. While it produces a classification, XGBoost does not give an indication of its certainty level regarding the result. Which is why it will be complemented by the results derived from the NGBoost engine; which extends the capabilities of the XGBoost by adding probabilistic predictions for the outcome, which means producing an estimated likelihood of the transaction being fraud. Instead of simply outputting the classification for the transaction, it gives a score of how big is the potential risk. Those scores will allow further flexible decisions compared to classical binary rules. Like a transaction with a high level of misclassification can be automatically blocked; while transactions with a moderate likelihood of being fake (based on the scores of the model) can be just flagged for manually checking.

Additionally, a notable feature of the architecture is the degree of explainability within the AI by the SHAP (SHapley Additive explanations). In most machine learning systems, where the algorithm used is complex, the decision making process cannot be explained back to the end user. SHAP (SHapley Additive explanations) can give this insight as it explains how each feature affects a specific prediction. This is done by looking at the value the feature has been assigned and if that feature has influenced the decision to be positive or negative. Going back to the fraud detection example, a fraud flag may have been determined due to altered values in the transaction, location and time. The system will be able to say that this transaction was predicated to be fraud based on the high value, location and time of the transaction. This is a highly desirable feature in bank and credit institutions where the value of the decision needs to be understood by the user.

The final stage of the architecture is the result presentation layer, where the output of the system is displayed to the user in a clear and understandable format. The system provides multiple pieces of information, including the fraud status, risk score, and feature-based explanations.

The last of the architecture is the final presentation of the result. The system shows the results to users through a well-presented output. The system shares with the user a few interesting reports, such as if the transaction is probably fraudulent or not, its score, and explanation features that the transaction is made from. All the pieces of information is displayed to the user in a lucid way through dashboards, charts, and other visual interfaces. This implementation increases the potential purpose of this research, which is to avoid human interactions. Returning raw data to users can be overwhelming. Therefore, this output is organized in such a way that users can decide quickly and effectively. The ability to interact with users and display results in a simple way is another merit of this system. Finally, the entire architecture is built on a guided flow that begins with data collection and ends with data visualization. The pipeline is interconnected to transfer data smoothly from one phase to another. Simultaneously, the design is modular so that each module is independent of others. This design permits other modules to be replaced or upgraded to higher techniques without damaging or affecting the whole pipeline. A memorable characteristic of this architecture is combining all techniques responsible for different issues encountered in fraud detection, such as data imbalance, high accuracy, explainability, risk probability. This combination results in a balanced system in terms of accuracy and interpretability. Moreover, the system is scalable over time, which is a vital consideration in part of the application that can not do the deep investments on budgets, hardware infrastructures.

In conclusion the design of this fraud detection system is an intelligent and comprehensive one that has done very well to incorporate modern technologies that can help solve real world problems. It is the next level, it doesn't just optimize for accuracy but also tries to eliminate issues surrounding data imbalance and uncertainties and the opacity of models. Its pipeline based model architecture is an addition to the already intelligent design as it ensures that each stage contributes to the system in a way that makes it overall very robust and reliable.

Algorithm of Proposed System:

Input: D: Historical dataset, R: Request

Output: Fraud prediction results / Batch analysis

- 1 Begin
- 2 Read D
- 3 if isImbalanced(D) then

```

4   D ← generateSyntheticData_CGAN(D)

5   end if

6   M1 ← train_XGBoost(D)

7   M2 ← train_NGBoost(D)

8   saveModel(M1), saveModel(M2)

9   initializeServer()

10  loadModel(M1), loadModel(M2)

11  E ← initializeSHAP(M1)

12  while true do

13    R ← receiveRequest()

14    if R.type = "single" then

15      sender ← R.sender_upi

16      receiver ← R.recipient_upi

17      amount ← R.amount

18      category ← lookupRecipient(receiver)

19      F ← generateFeatures(category, amount)

20      F ← preprocess(F)

21      P ← predict(M1, F)

22      if category = "Blacklisted" then

23        P ← max(P, 0.95)

24      end if

25      if  $P \geq 0.75$  then

26        risk ← "HIGH"; action ← "BLOCK"

27      else if  $P \geq 0.45$  then

```

```

28     risk ← "MEDIUM"; action ← "VERIFY"
29   else
30     risk ← "LOW"; action ← "ALLOW"
31   end if
32   S ← computeSHAP(E, F)
33   explanation ← getTopFeatures(S, 5)
34   sendResponse(P, risk, action, explanation)
35 else if R.type = "batch" then
36   T ← loadCSV(R.file)
37   T ← preprocess(T)
38   P_list ← predictBatch(M1, T)
39   S_list ← computeSHAPBatch(E, T)
40   for i ← 1 to length(T) do
41     if P_list[i] ≥ 0.85 then
42       risk[i] ← "HIGH"
43     else if P_list[i] ≥ 0.60 then
44       risk[i] ← "MEDIUM"
45     else
46       risk[i] ← "LOW"
47     end if
48     explanation[i] ← getTopFeature(S_list[i])
49   end for
50   stats ← computeStatistics(risk)
51   sendBatchResponse(P_list, risk, explanation, stats)

```

52 end if

53 end while

54 End

3.2 Modules in the proposed work

For the proposed UPI fraud detection system, it is designed as a flexible, modular architecture which can be easily updated, extended and refined over the years. Instead of a monolithic complex system, the fraud detection system is decomposed into smaller modular components with well-defined APIs, integrated to form a complete solution. Each module is used to perform a specific task without impacting the other modules, while making it easy to incorporate new modules or replace existing ones. This type of design allows for the addition of new features such as real-time fraud detection or state-of-the-art machine learning techniques. The architecture consists of several modules that deploy a variant of a common service-oriented architecture. The front-end module is the user interface, and the back-end module performs feature extraction. In order to deal with the “red-tape” problem of imbalanced fraud data, a CGAN based dataset balancing module generates realistic fraud samples and feeds to the machine learning module. The machine learning module conducts transaction classification and fraud scoring. Finally, the result presentation module summarizes the results, such as fraud decision, risk score, and reasons, and presents them in an elegant and intuitive way. All modules are tightly integrated to communicate data effectively from one to another. This well integration ensures end-to-end processing of each UPI transaction from beginning to end without disruption.

3.2.1 Frontend Module

The frontend module is the first step in the system, acting as the user interface to the fraud detection system. It must provide a means for transactions to be entered into the system, validate the data within the user input, and hand it over to the next stage in the system to analyse. User data can be entered either by uploading a CSV file or by manually entering each individual transaction. The frontend then needs to check that the transaction amount, date/time, merchant name, and other information needed are all present, and set correctly to prevent any problems on the downstream processing. The frontend was constructed using a number of trendy web technologies, using React.js together with HTML, CSS and JavaScript, in order to create an attractive, and responsive user experience that was easy to access and simple to use.

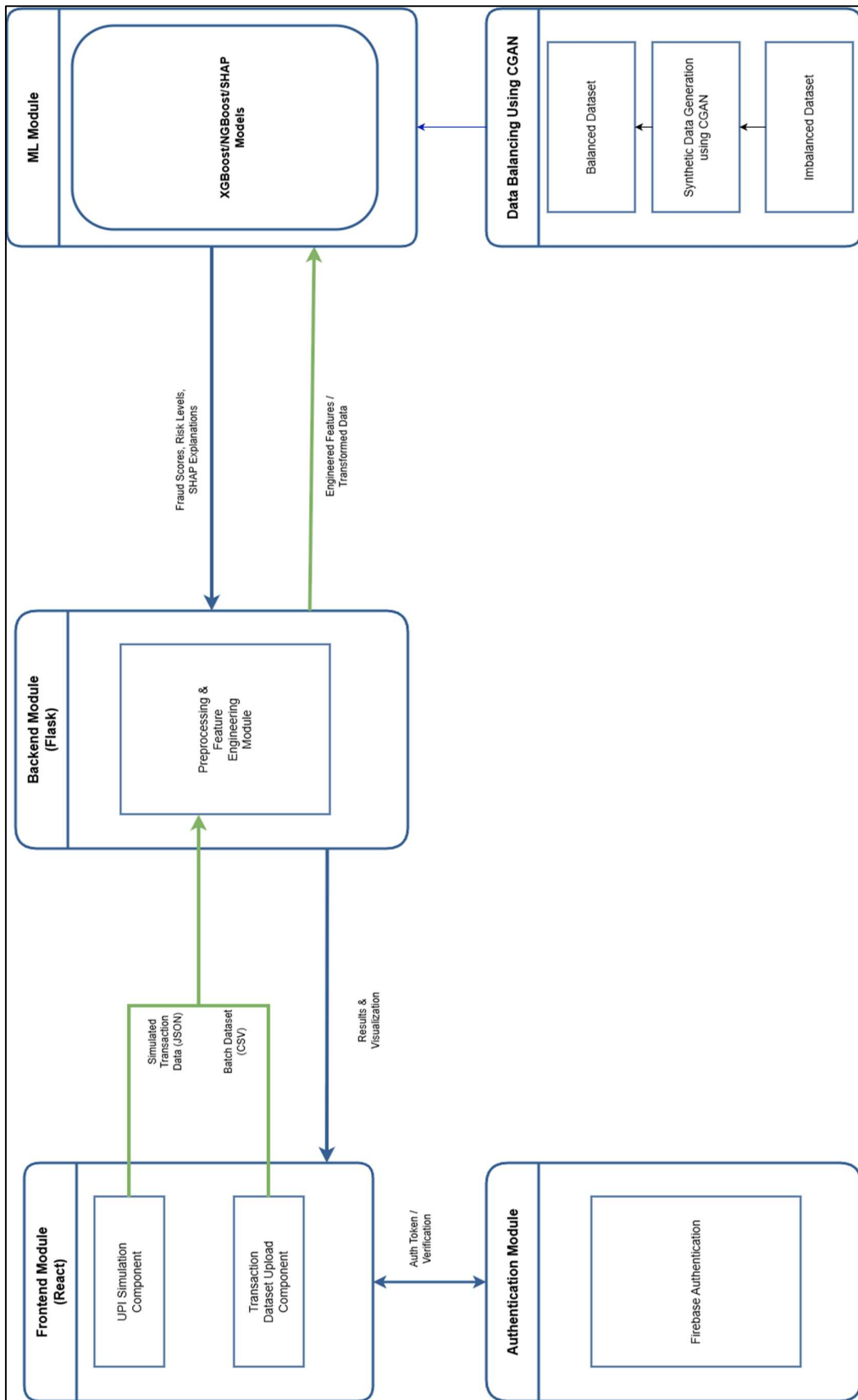


Figure 2: Module Connectivity diagram

This offers a pleasant entry point for the user to enter details, and view the results without having to worry about the technological details running in the background. The front end communicates with the backend using JSON formatted API, ensuring the efficient transfer of data between the parts. Security is important for the frontend as well, by using mechanisms such as tokens the data being sent from the user to the server for processing is secure.

3.2.2 Backend Module

The backend module is the heart of the system. It is responsible for most of the calculations on the system. It takes the transaction data from the front end and makes it ready for the further processing, it also cleans the data, by removing any missing data, removing any invalid or useless data and also making sure the data is easily accessible. Apart from the cleaning other functionalities the backend performs are, feature extraction, for extracting out the valuable information such as the date, time, the number of transactions in a day, the change of in write of the transaction, etc. These are the preparation to get the data ready for machine learning algorithms. The backend for our system is developed using Flask, a light weight Python web framework it enables easy management of web-based systems. It extracts the data from the service modules like the pandas and NumPy. The main task that the backend module does is managing communication between various modules, through this data flows from front end to the machine learning modules. The backend provides a security mechanism, where the user is identified, managed and then allowed access. It also checks the path of the data from the front end to the prediction engine before presenting the user with a result. It makes sure that the data is valid else it does not show the result. It is flexible and scalable in design, so that new features, better data processing techniques or models can be added without effecting the system as a whole.

3.2.3 CGAN Dataset Balancing Module

The CGAN module is targeted for solving one of the biggest issues in fraud detection, the class imbalance. In most of real-world datasets, fraud class is very small in comparison to non-fraud class. This makes it difficult for machine learning algorithms to learn the fraud class correctly, because they get very few fraud instances. The CGAN module is proposed to overcome this challenge by learning the characteristics of real instances and generating realistic synthetic instances. Conditional Generative Adversarial Network (CGAN) is a generative network with an embedded classifier. It attempts to learn the distribution of data/transactions and generate new instances with similar properties. CGAN network consists of a generator and a

discriminator. The generator network takes noise vectors along with conditional information like transaction time, amount and merchant details as input and generates synthetic fraud transactions. The discriminator network takes both real and fake transactions as input and learns to classify whether the transaction is real or fake. The generator and discriminator networks are trained simultaneously. The generator network tries to generate convincing transactions such that the discriminator cannot tell the difference between a real and generated transaction. The discriminator then becomes even better at classifying true and generated instances, while the generator becomes more successful at generating convincing instances. Once the CGAN network is trained, it generates realistic synthetic instances of fraud transactions for the given conditional information. These synthetic instances are combined with the original data set, which results in a balanced data set and improves the quality of classification models trained on this data. The synthetic instances are more reliable than existing over-sampling techniques such as SMOTE because of their genericity and contextual information. This approach improves the fraud detection capabilities of the system.

3.2.4 Machine Learning Module

The machine learning part of the system does an important job. It searches for transactions and tries to estimate how risky they are. XGBoost is the component that searches on the data for patterns. It is extremely good at searching numerical data for hidden relationships. This is done by creating a number of decision trees, each of which looks at the mistakes made by the previous tree and tries to make a decision that is better. The XGBoost component is tuned to make sure the model does not get so good at analyzing data that it forgets how to analyze data it is never seen before. NGBoost is the component that attempts to estimate how risky a transaction is. It can go beyond a simple yes/no answer, and provide an estimate of how confident it is about a decision. This is extremely useful for tasks related to money. The system also uses SHAP. This tool provides an easy way to understand the rationale behind a transaction being marked as fake, and shows how much each individual attribute contributed to the final decision. This capability allows the system to be highly dependable and trusted. The machine learning component enables all of this by providing a method of analyzing data, estimating risky, and explaining choices.

3.2.5 Result Presentation Module

This is the module that displays the output of the system in a visual appealing, easy-to-understand way. It takes the output from the machine learning module such as the fraud class,

the probability score, SHAP values, etc. and displays them in dashboards, charts and visually appealing dashboards. Instead of the user searching through all the output, he/she can see the state of each prediction, if a transaction is bad, for how much risk, the features that lead the system to classify that transaction as illegal. The result presentation module will use charts and graphs to indicate certain trends and alerts to help the user make decisions and to lighten up the system. The result presentation module will also make sure that the output is well structured so that the user can understand the output and other systems can use the output.

3.2.6 Module connectivity diagram:

The next module connectivity diagram (Figure 2) displays the interrelation of all the components of the system. It represents flows from the frontend to the backend, then preprocessing, machine learning and generation of results.

The Proposed AI system in Figure 2 is to detect the fake UPI transactions in real-time. It is a collaborative system that working with each component to perform different functions. This system has a user interface that provides users the ability to enter data (or) upload a file that contains a whole bunch of transactions data. It then analyzes the data and predicts whether it is fake (or) real based on its prediction and interprets the reason for such prediction. Detected Frontend component firstly which is user interactive component of this system, built on the React platform. This component has two different functionalities provided to the user: one for simulating fake transactions for testing purpose, and the other to upload a file with transactions data, which is then sent to backend component. Backend component works as brain of this system, which receives data and analyzes it to predict whether the data is fake (or) real, and interprets the reason prediction made. The communication occurs between these two components using JSON, CSV, etc. Formats.

Have a special authentication module that ensures only authorized users can use the system. This module leverages Firebase Authentication to create users, allow them to log in, and handle their sessions. When a user logs-in they receive a token which must be included in every request they send to the system. This way the system is protected and only authorized users can gain access. The backend is the central part of the system. It is built using Flask. It receives requests from the frontend, validates that they are authentic and passes them to the relevant part of the system. An important part of the backend is the preprocessing module, which receives the raw data from the transactions, cleans it, replaces missing values, and standardizes it before passing it to the machine learning module.

Once the data has been pre-processed it is sent to the machine learning module, which is the 'life' of the system. It employs 2 models (XGBoost and NGBoost) for classification of whether a transaction is real or fake. XGBoost is a high-performance implementation of gradient boosting and it is especially good at detecting regularities within data, while NGBoost is capable of providing not only a classification but also an actual probability to the prediction. Combining the power of both models the system is able to accurately predict the class of a transaction. The system goes beyond the simple binary decision of whether a transaction is fake or not.

It also explains to us that why it made the decision. This explanation is achieved by an explanation method known as SHAP, which considers each feature's contribution to the decision. This is very critical in systems where user want to understand why a decision was made. Another key component of the system is the data balancing component, which is an approach referred to as Conditional Generative Adversarial Networks (CGAN). This component is used to prevent the system from being biased to particular type of transaction by creating transaction data with the same distribution of real data but not the same. This facilitates the system to learn from a balanced data distribution which improves its performance to some extent. To conclude, the Proposed system is a solution to deter fake UPI transactions. It combines modern web technologies, secure authenticators, advanced machine learning models and sophisticated machine learning models and sophisticated explanation methods to build a system that is not only accurate but transparent as well. The Proposed system contains several components each with unique functionality which work in unison to build a secure and safe system. The system is great at detecting fake transactions but it also explains to you why it believes that a given transactions is fake which is very important to users of the system. The Proposed AI system can be used to fight fake transactions in many ways.

3.3 Analysis and Design through UML

Analysis and Design by UML (Unified Modelling Language): is understanding and a system design, Before you actually putting a system together. Analysis phase. It helps us identify and specify the system requirements by models that capture the content, structure and interactions occurring within the system, typically presenting these with use case diagrams. Design phase: UML provides internal structure of how the system should be designed using class diagrams, Sequence diagrams and activity diagrams. The benefit of UML is providing a design for the system that can be easily understood by developers.

Use case diagram:

The system in Figure 3 can be divided into two major roles: user and system administrator. Users can either input transaction information or input a UPI ID and the system will check the transaction and output a fraud risk score with reasons. This is an end-to-end system, which enables better accuracy and confidence since the user can explicitly know why the system categorized a transaction as risky or safe. We use both XGBoost and NGBoost Models.

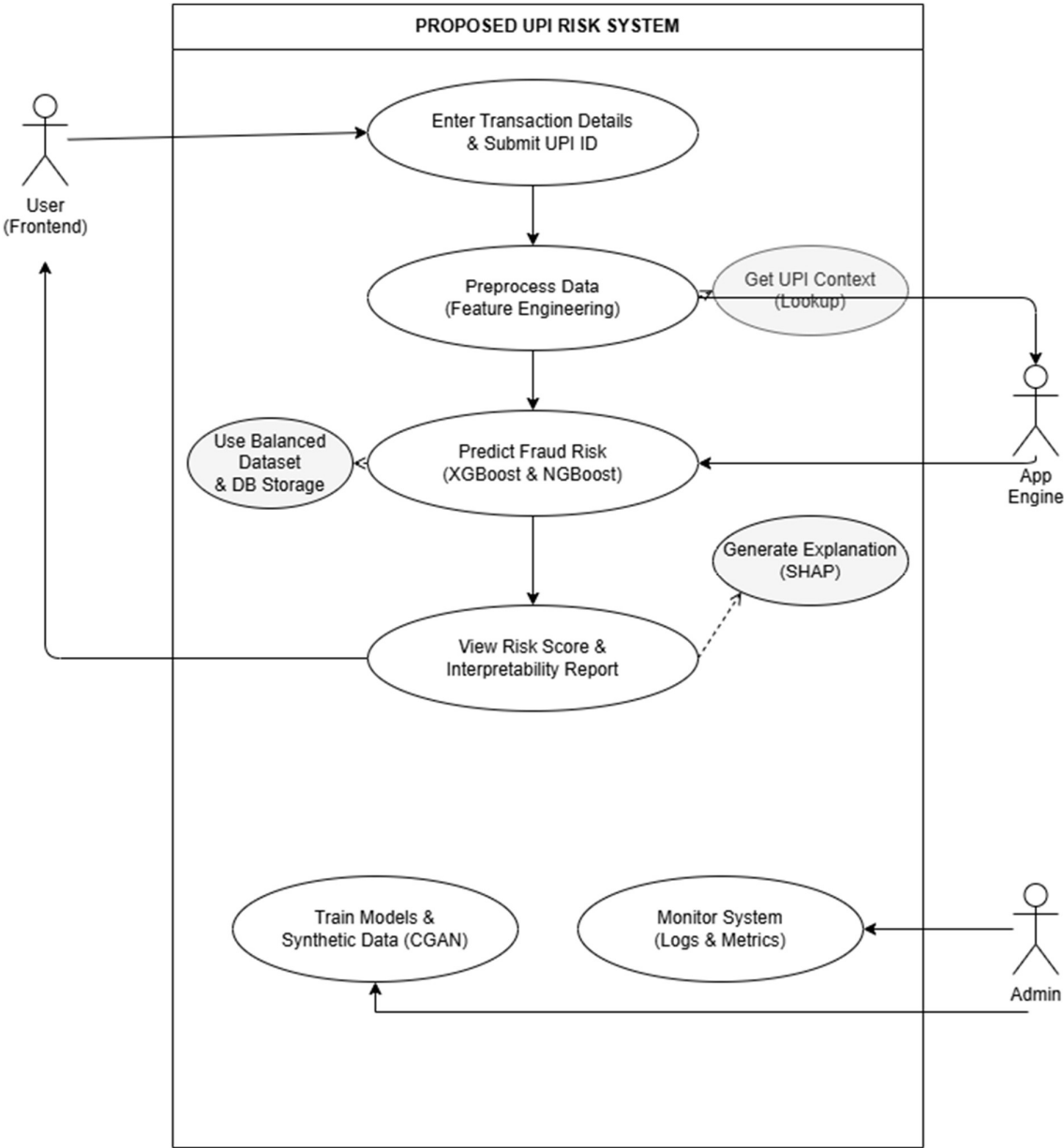


Figure 3: Use case diagram

Combining them we get a more accurate risk score. Results explanation is used to improve the transparency of the system by indicating the main factors that have led to the final decision. This is done using SHAP (SHapley Additive explanations). Handling imbalanced dataset is another important part of the system, as fraud detection is usually based on class imbalance datasets. Since class imbalance is a problem in this case as well, CGAN is used to generate artificial data and balance the dataset. The role of the system administrator is important with respect to data generation, training the model, updating the database of the UPI and monitoring the performance of the system.

Overall the Proposed system integrates user interaction, safety to data processing and prediction intelligence (predictive model) and interpretability in a single system to detect fraud. Figure 3 shows a comprehensive system generated which is modular, simple and for production purpose it can be scaled, maintained, modelled and works efficiently.

Class Diagram:

The class diagram shows the internal structure of this system, displaying different classes for this system and their relationship.

The class diagram of the proposed system shows how the different parts are integrated together in a very simple way as each class does one simple task. As we see in Figure 4, the Proposed System class acts as the brain and the coordinator of the system. It just organizes the input and the output of the different classes. It loads up trained models from the database, takes transaction input, predicts whether transaction is risky or safe. It then takes the output from the prediction class and organizes the next step accordingly. It doesn't perform anything by itself. Aggregator class as the other two classes need a lot of data before they can make predictions. The feature aggregator is the class which integrates and extracts all the required features from the detailed data collected. It cleans, organizes, and converts the data into the right format so that the prediction models can predict without any error. The system also has the UPI Database class which takes information about the users, merchants, and suspicious UPI IDs from the database and makes the data more detailed. It enriches the transaction data with additional information like past transaction history and IRS risk value, and trust score of the customer, and adds it to the final prediction input which improves the performance of the machine learning models. As the data stored in the database is a JSON file, it is very easy to update or add data in the database. Also, to account for the lack of enough number of fraudulent

transactions, the system has the CGAN Model class which generates fake data similar to real data and generates balanced dataset.

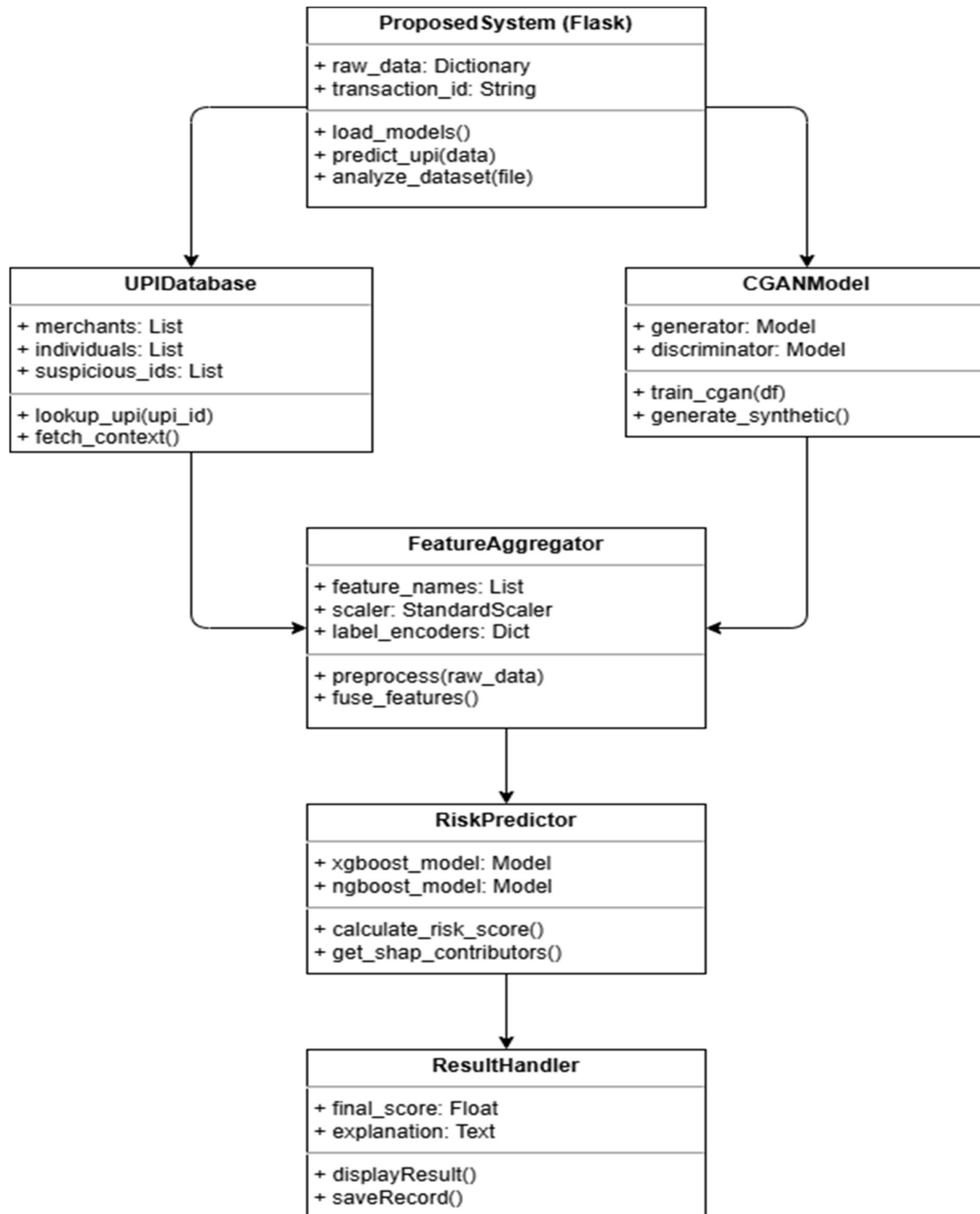


Figure 4: Class Diagram

The actual decision making is performed in the Risk Predictor class which takes the input on the prediction class in the form of features as well as meta data and predicts whether the transaction is risky or normal. It uses machine learning models such as XGBoost and NGBoost to predict whether a transaction is risky or not as well as derive an explanation as to why it has taken a decision. This makes the system more transparent. To work, it depends on the Feature

In general, this design where every class is working as member of a team is efficient for a big system. Proposed System class is orchestrating every page and taking decision, Feature Aggregator is collecting and normalizing data, Risk Predictor is taking decision, UPIDatabase is supplementing the data, and CGAN Model is enhancing data quality. As can be seen from Figure 4, this separation of responsibilities is clear, scalable, and robust. This is a reasonable design choice for a real-life financial system requiring both accuracy and understanding.

Sequence Diagram:

The sequence diagram is a diagram of how processes work in the transaction. The diagram displayed is the Figure 5, showing requests flow from the user to back-end, back to the machine learning model, and back to the user. The time flows down the page so the diagram is easier to understand. The system as specified (Fraud Detection System) demonstrates how various parts come together in order to process a UPI transaction in a very intelligent yet efficient manner. As explained in figure 5, the process starts when a frontend user sends an API request through Flask, which contains the main details of the current transaction such as from, to, amount etc. The system receives this request and begins processing immediately. The first step towards processing the transaction request is to locate additional information about the receiver by searching the UPI database, here the system would study previous transactions whether successful or not, examine whether the account is genuine, and check for any other suspicious activity. This would give the system additional context of the transaction and aid decision making and detection of anomalies. Once the location of the receiver has been learned, the system further preprocesses this information which gets sent to the Feature Aggregator, this stage ensures that the data is cleaned and arranged in the correct format for advanced analysis, the text features are converted from words into existing term frequencies and the numerical data is scaled and transformed to be in an ideal range for the models, this step ensures that the analysis is efficient and excellent accuracy can be achieved.

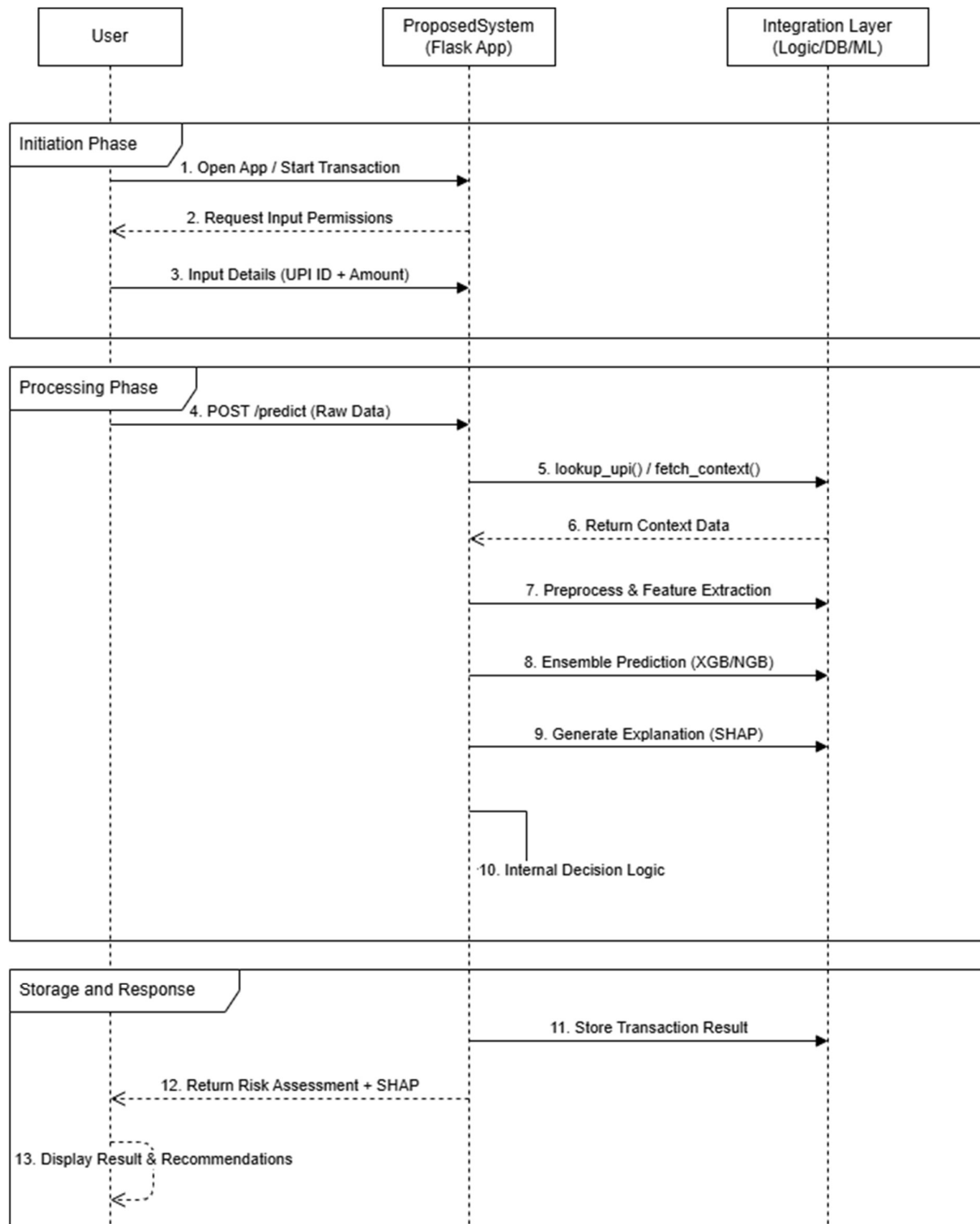


Figure 5: Sequence Diagram

Thus, after data preparation the system proceeds to analyze the transaction request and come to a conclusion. Using the combined models of XGBoost and NGBoost, trained to identify the key imbalances that could identify a fraudulent transaction, the system analyzes the data and makes an effective prediction. As shown in the figure, the expert model (XGBoost) plays a

monumental rule in outperforming the weak base model (NGBoost) and increases the efficacy of the system drastically. The combined models are followed by an explanationer which identifies the contributions of each feature (SHAP explainer) and provides information about the result which makes the system very transparent. Based on the analysis, the system then determines what to do with this transaction, whether to block it, ask for more details or to confirm and complete, and delivers the output to the user.

To sum up, as from Figure 5, the whole Fraud Detection System acts as a very effective pipeline getting the data in supplementing, pre-processing, and analyzing it, explaining its result. This systematic data flow ensures accuracy, openness, and efficiency of the system. It is a robust and efficient tool safeguarding UPI transactions, so that users could enjoy a safe and comfortable digital payment experience.

Activity Diagram:

Figure 6 shows the whole process of the system to be followed step by step. The diagram describes how the transaction is to be carried out from the input stage up to the final decision stage, such as checking for validation, analyzing the data, and generating the results. It also shows different paths possible, which is, whether a transaction is fraud or not, whether it is verified or not, or whether it is approved or not. The diagram illustrates the intelligent system which analyzes the transaction as it happens and safeguards the UPI transactions. It guides us through the process, as in Figure 6. As soon as the user presses the transaction button, the system captures the basic details such as sender, receiver and amount and uses these as input for all the following checks. Once the transaction is initiated, the system jumps immediately into receivers UPI ID and finds out if the receiver has any information stored. If it does find any, it brings these into the flow. Otherwise, it creates a default set of information and walks forward. The information can be about previous transaction behaviour, manner of trading and trustworthiness of receiver. This helps the system to understand the nature of the transaction, whether it is normal or not. Once the system has found out and collected the information, it prepares it for classifier or decider. To do this, it sorts and converts the information to a learned structure. While doing this, the system test the amount of evidence in the data, so that it can make the most accurate decision; it then implements the classifiers or the (bag of models) of XGBoost and NGBoost. As the names suggest, XGBoost helps the system to unearth which models succeed most, while NGBoost helps to estimate the confidence of this prediction. This

all contributes to the system a robust assessment of how risky the transaction might be, and it then provides an explanation of the weights of the decision.

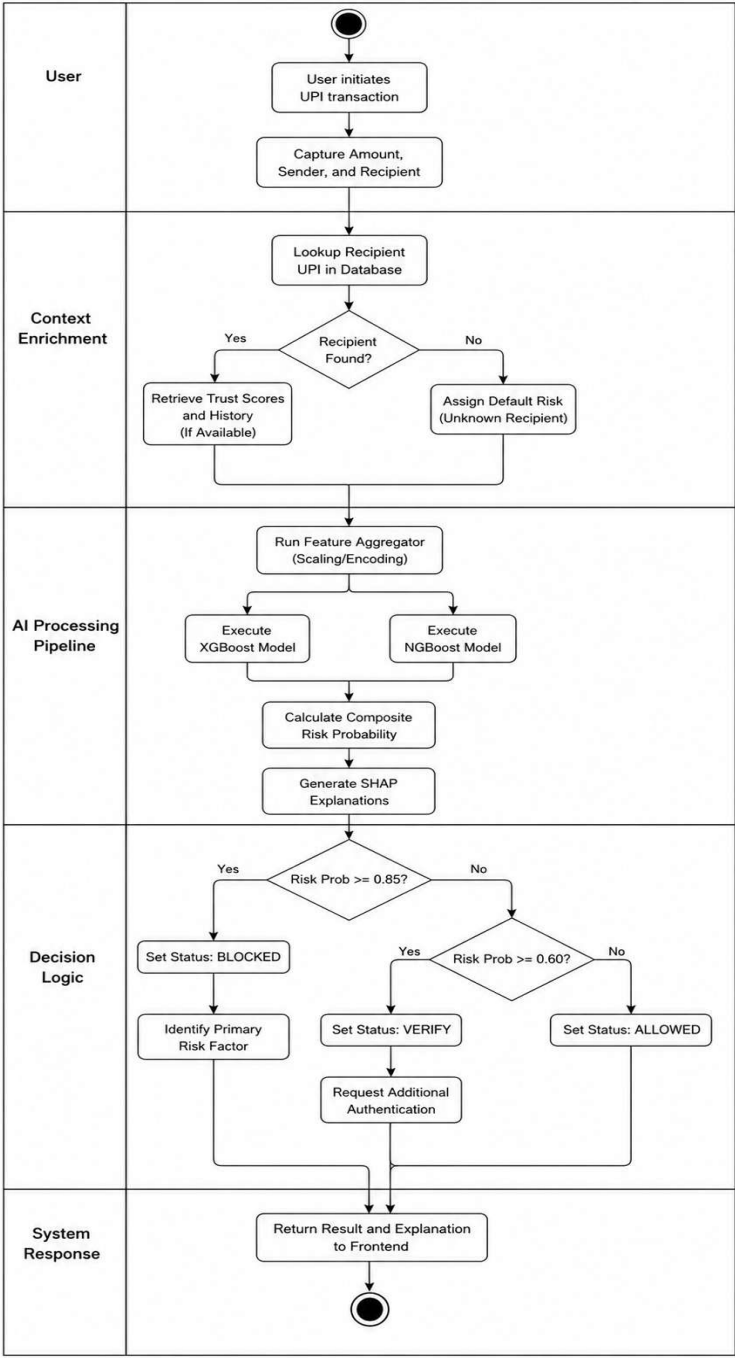


Figure 6: Activity Diagram

If the prediction is suspicious, then the system forges the transaction but if the system is unsure, then it requires additional verification through user confirmation, one-time password etc. If the transaction is not suspicious it moves on freely. This is communicated to the user along with the evidence of the weight of the prediction.

3.4 Data and Control Flow

From the system, the Level 0 Data Flow Diagram (Figure 7) shows the interaction between the user, system and model store at a high-level, while reducing the level complexity. The user interacts with the system providing CSV data sets or UPI transaction data as input. The System reduces the data and analyzes it, computing the risk score, visualization, alert. It forwards this to the user. It internally forwards the reduced form of data to the model store, where it gets trained. The trained models are then brought for producing a prediction or probability-based risk score.

Diagram below shows the ‘flow’ of data and how it is represented in the sections of the system. Loads ‘transaction data’ from the user/client to pre-processing, model, analysis and final output. The data is handled so that each part of the system pre-processes and passes it to the next part. Used to understand the task better. The Level 0 DFD enables to explicitly identify the scope of the system and the major data processing requirements. It helps to develop an understanding of flow of data between various components without internal details. The diagram is also useful to the developers and other associated projects teams when designing systems and later implementing that system. It also provides basis for further elaboration by developing lower level DFDs for in depth analysis.

Going a level deeper, Level 1 Data Flow Diagram (figure 7) illustrates the internal flow within the system into several interconnected processes. The flow begins with the admin providing an unbalanced dataset, which then is sent to a data balancing module, which applies CGAN to generate an improved balanced dataset, which then is stored and sent to the model training section of the system, where a Machine Learning algorithm such as XGBoost or NGBoost is applied to build predictive models. These are then stored and used during real time fraud analysis for the system. The diagram also elaborates more on the user interaction flow in a more practical sense. It shows how the uploaded transaction data is sent from the frontend interface to the backend (Flask API, classifying engine, etc). After the preprocessing step, it is sent to the fraud engine, which evaluates the uploaded data by using the trained models and

model-explainability techniques to generate the fraud score, risk level, and interpretability information.

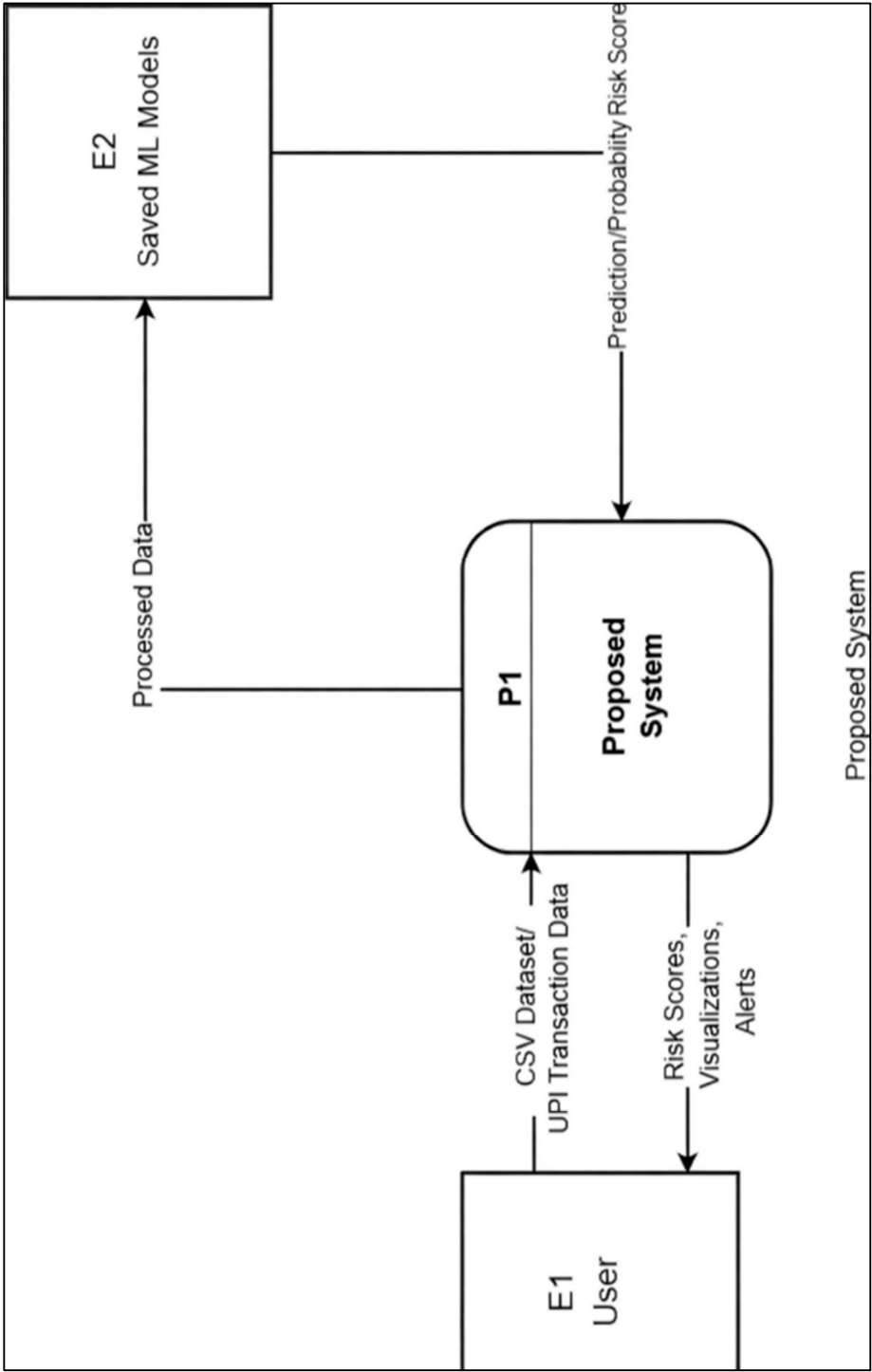


Figure 7: Data Flow (Level 0)

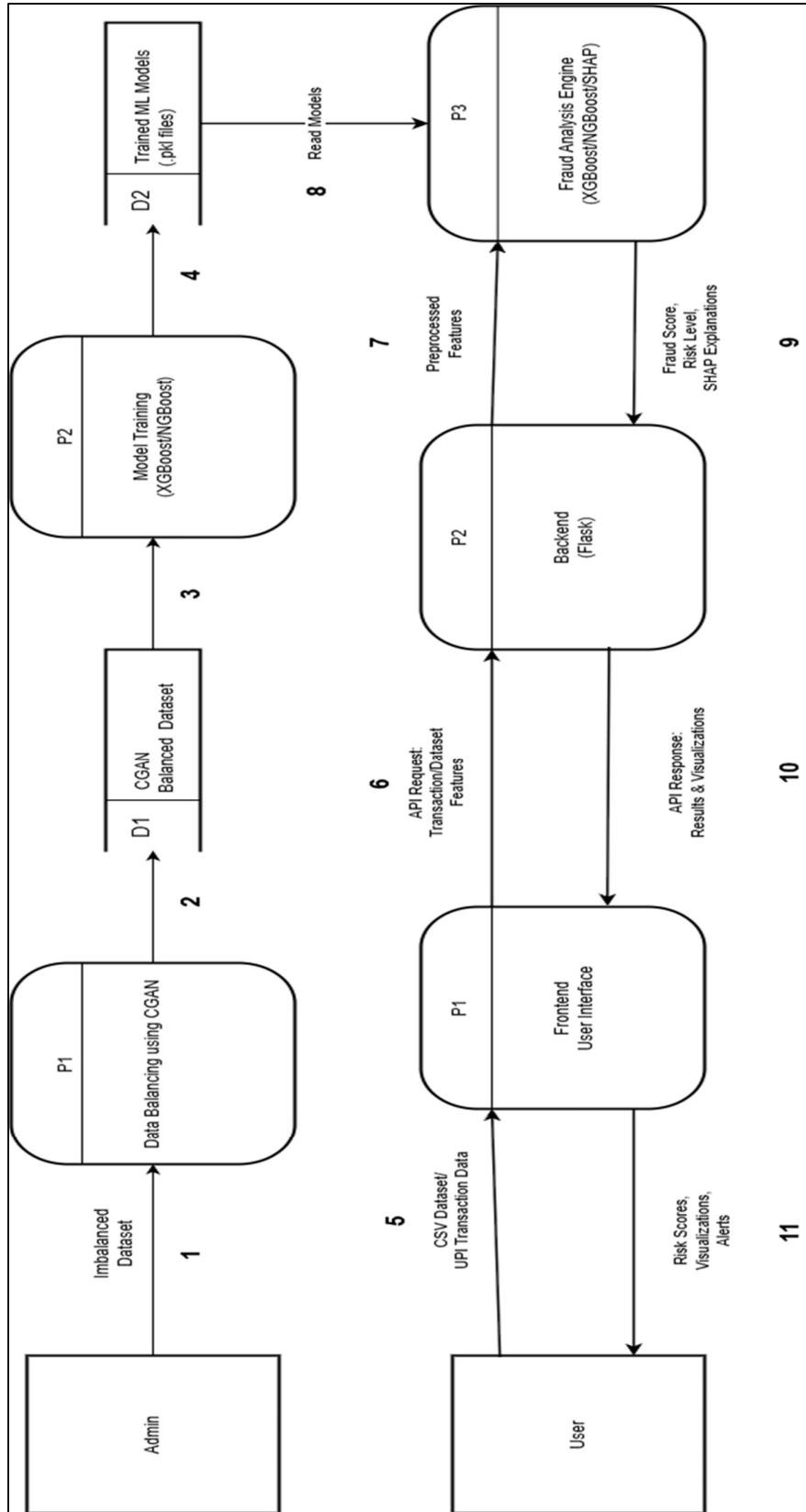


Figure 8: Data Flow (Level -1)

Control Flow Diagram:

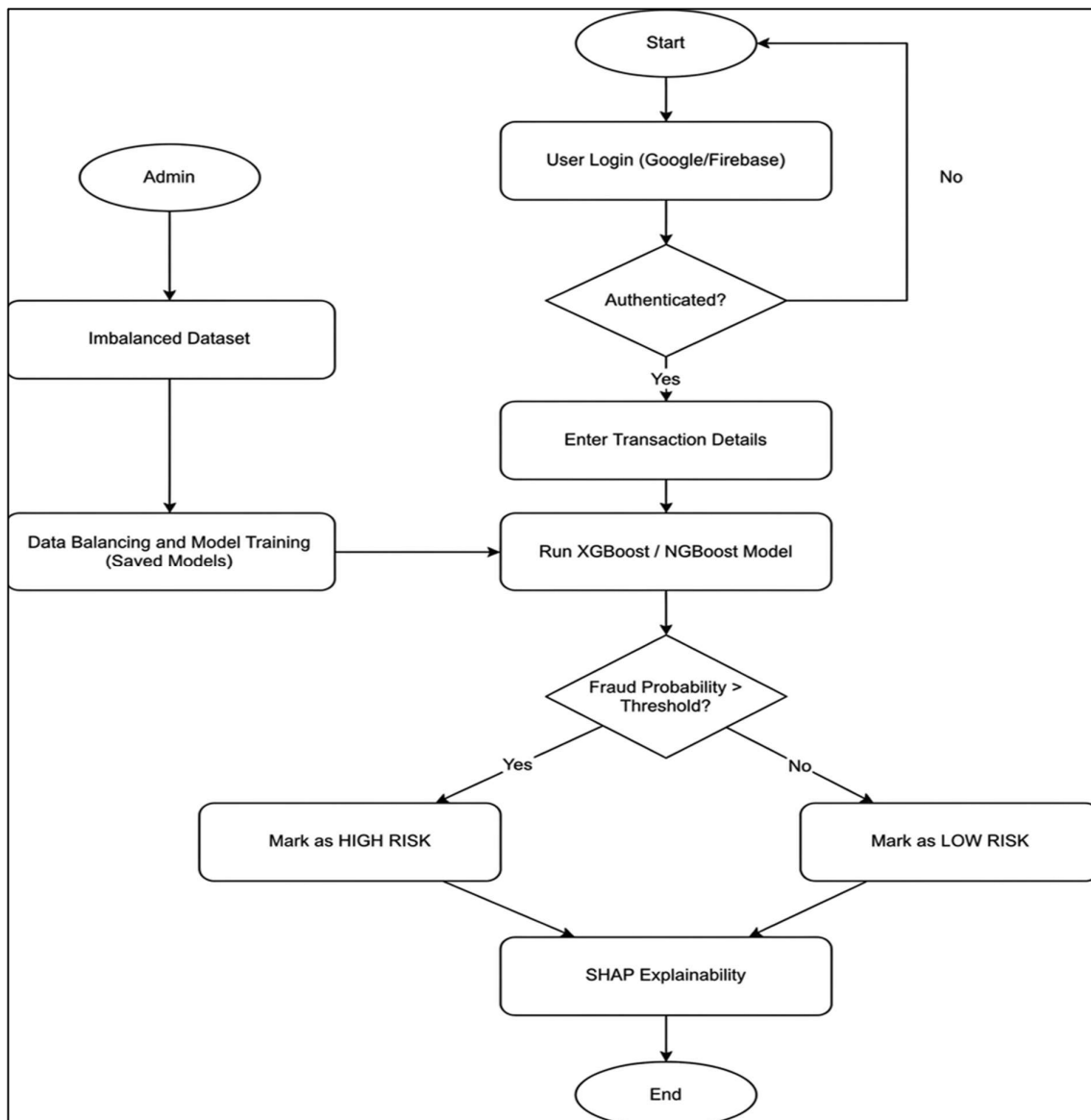


Figure 9: Control Flow

The flowchart as shown in Figure 9 reveals the end to end working of the system in a linearized form easy to comprehend and follow along. The input to the system is provided by the user by logging into the system during the login phase by authentication methods like google, firebase or any other third party source where in only authenticated users are allowed to log into. In the login phase if the login is unsuccessful then the process recomposes itself back-to login phase or else it proceeds to the input provided in the form of transaction details which will provide input to the fraud detection pipeline or process whereas on admin side a given imbalanced data set is provided in the train data, the data is passed through the data balance techniques and

model training phase where in the trained models like xgboost or NGboost etc. Are saved for better prediction accuracy. The trained models are thus fed to the fraud detection pipeline where in a transaction provided as input then is passed through a kafka server or any message passing medium into the server where the prediction analysis is done by providing the trained models to an application. The test data analysis which is passed on to server-computation of fraud probability and passing on the crime threshold (above or below threshold) to be classified as a high or low threat transaction and finally providing the output as an output only after passing through the SHAP-Explainability to generate features I will show how the resulting output is interpretable in nature.

3.5 Requirements Engineering

The requirements engineering defines the Required System (what needs to do, how well it is in preventing fraud). It contains the functional requirements (the main tasks to be completed by the system) and the non-functional requirements (the qualities of the system for proper operation and ease of use).

3.5.1 Functional Requirements

Following system is proposed which can identify fraudulent transactions and also provide effective insights to the user. Firstly, the system should enable the user log in using various authentication methods such as google, firebase etc so that end-user can log in and use the functionalities. Post login, user should be able to upload values like upi transaction data, export .csv file, dataset csv file etc. as their input and can pass it to the system. The data is then forwarded to the backend where it is pre-processed (divide it into data, normalize it, extract features, handle missing values) then can be passed to the trained models like xgboost, NGboost etc. which will give a probability of frauds in the transactions and based on the value (high risk, low risk), the transaction will be labelled. The system also should allow the visualizations, alerts, risk scores to be made available to the user. The system can also take advantage of the SHAP so that the user can get an insight into the explanation of whether the transaction is safe or fraud. From admin point of view, the system should allow access to the datasets, handle imbalanced data, handle data with CGAN, train models, save all models for prediction etc. The coordination between rest of the system should be seamless.

3.5.2 Non-Functional Requirements

Along with the basic functional capabilities the Proposed System also need to comply to a number of performance related and quality factors. Prediction result has to be provided fast, ideally just as a background transaction checking process or within a second or two. The prediction techniques need to be scalable, scalable to large number of new users and a large number of transactions in order to prevent a performance bottleneck. Since the system involves a lot of financial and private data, it needs to be secured against unauthorized access and data loss. The system should be reliable in providing accurate prediction results, and be operational with the minimum down time. It should be easy to operate with a user-friendly interface. It should be easy and flexible to be maintained. Being able to update models, fix bugs, extend features easily. Finally the system should be transparent and able to explain reasons behind decisions.

The system should allow for real-time processing in order to process large amounts of transactions as they happen. The system should also be highly available and fail safe. The system should be partially upgradeable so that future technology advances do not require a major redesign of the infrastructure. There should also be mechanisms in place to allow logging and monitoring of the system. It is also important that the system complies with Data Protection regulations.

3.6 Software and Hardware Requirements

Table2 & Table3 specifies the system hardware and software requirements that brings out the system and tool specifications needed for the development of project.

Table 2: Software Requirements

Serial NO	Software	Purpose
1.	Python 3.x	This serves as the primary programming language used to develop the entire analysis and prediction system.
2.	NumPy, Pandas	Used for efficient data manipulation, preprocessing, and handling large transaction datasets.
3.	Scikit-learn	Used to build, train, and evaluate machine learning models for fraud classification.
4.	TensorFlow / PyTorch	Required for implementing and training deep learning models such as CGAN.

5.	XGBoost	Used for accurate classification of fraudulent and legitimate transactions.
6.	NGBoost	Provides probabilistic predictions and risk estimation for fraud detection.
7.	SHAP	Used for explainability by identifying feature contributions in predictions.
8.	Flask	Backend framework used to handle APIs and system integration.
9.	HTML, CSS, JavaScript, React	Used to design and develop the frontend user interface.
10.	Jupyter Notebook, VS Code	Used for coding, experimentation, and model development.
11.	MySQL / MongoDB	Used for storing and managing transaction data.
12.	Matplotlib, Seaborn	Used for data visualization and analysis through graphs and charts.

Table 3 Hardware Requirements:

Serial No	Hardware	Purpose
1.	Processor (Intel i5 / i7+)	Required for efficient model training and execution.
2.	RAM (8 GB / 16 GB)	Used for handling large datasets and performing computations.
3.	Storage (256 GB / 512 GB SSD)	Provides fast data access and improves system performance.
4.	GPU (NVIDIA - Optional)	Accelerates deep learning model training such as CGAN.
5.	Internet Connectivity	Required for dataset download, updates, and integration.

CHAPTER 4

Results and Discussions

4.1 Description about Dataset

The data set used here is the UK Handing Fraud data set from the IEEE-CIS Fraud Detection Data Set. This data is the standard database that is utilized in the construction and assessment of systems regarding financial fraud detection. It consists of nearly 20000 real-world transaction records, of which some of the activities are legitimate while others are counterfeit. This data set is distinct since the imbalance present means that around 90% of the data is valid, leaving only very little of the data to be forgeries. As this is the case, the machine learning algorithms that are used to train the models have to then learn to have predictions that lean toward the transactions being legitimate, but this leaves untrained the ability to detect fraud. The problem is that there needs to be a specific way in which the data is prepared, such that the model can distinguish between the transactions and decipher whether or not said activity is suspicious. One of the features of each transaction is that it will include additional metadata that can help in contextualizing the commerce. Many factors such as transaction volume, when it was performed, what store it was conducted at, and what location it was carried out at, as well as the behaviour history of the consumer, are crucial to acquiring the inconspicuous differences between what is normal and what is suspicious. It comprises both categorical and numerical data, which will mainly include the time and sum of the transaction, and whether or not specific goods were imported or exported. The data also must be prepared for use in machine learning models since the models require the data to be reformed into numerical inputs, so the combination of the features will have to be converted into a number, and then scaled.

Before being used for training purposes, the original data set is being transformed through a number of preprocessing and feature engineering processes. New features are being engineered such as mean amount of transaction, frequency of transactions and behavioural patterns of users providing important information about users' habits of spending. Because the data set may contain a lot of missing values and noise, strategies for data cleaning like imputation are applied as well as Outliers analysis to decide if a specific value is actually some sort of abnormalities or fraud. All these processing, transformation and engineering efforts aim to generate a data set easy enough for the computational models and robust enough to be really representative of the real-life scenarios. This is the very reason why techniques like generative models for data

balancing, genetic algorithms or ensemble learning are being adopted. To conclude, the IEEE-CIS data set is greatly appropriate for this project as it contains all the relevant real-life information and bear the potential to develop a reliable fraud detection system.

As to overcome the problem of data imbalance, techniques like SMOTE (Synthetic Minority Over-sampling Technique) or under-sampling methods can be used. Feature selection techniques are also important to remove the less relevant attribute and further alleviate the curse of dimensionality. Cross validation techniques can be used to prevent the model over fit. More over to measure the effectiveness of the fraud detection system, it is vital to report the measures like precision, recall, F1-score and ROC-AUC etc. Finally, you cannot exactly one shot build the accurate model as the fraud situation can evolve with the evolution of fraud patterns so continuous model updating and monitoring is needed.

Table 4: Dataset Details

S.NO	Attribute	Description
1	Transaction ID	Unique transaction identifier
2	Transaction Amount	Monetary value
3	Transaction Time	Time of transaction
4	Merchant ID	Merchant identifier
5	Merchant Category	Type of business
6	Location	Transaction location
7	Previous Transactions	Number of past transactions
8	Average Amount	User's average spending
9	International Flag	Domestic / International
10	Fraud Label	Target variable

Dataset Challenges

Based on Table 4, the dataset in proposed System contains a number of real-world problems that consists of the difficulty in detecting fraudulent transactions. The first issue is the highly skewed dataset. The imbalance between classes turns out to be one of the most significant problems in fraud detection. In the dataset most of the transactions are fraud-free and very few transactions are with fraudulent activity. The unbalanced data distribution can bias the machine learning models towards predicting only normal transactions, which affects the performance in

uncovering fraud transactions. There are also noisy, missing data in the dataset that can affect the prediction performance if the problem is not considered in the design stage. Proper handling of missing data is crucial for prediction accuracy so it is important to apply various pre-processing methods in the system such as data cleaning, imputation and normalization to generate consistent dataset. The second difficulty is the overlap between fraudulent and genuine transaction patterns. In many fraud cases, the fraudulent activity tries to look like normal transaction, leading to difficulty in between normal and fraud instances classification problems. The last problem is the non-stationary class distribution in the data. Fraud behaviors evolved over time to a different attack type, and the model continues to evolve continuously to cope with these changes. In summary, the data set is successful as reflecting real financial scenarios which occasionally include a wide range of severities of noise, blank-field anomalies and confusing trends. The types of degeneracies encountered in practice indicates the importance of sophisticated pre-processing pipelines and ensemble learning for the modelling.

4.2 Detailed Explanation about Experimental Results

In the Proposed System, how to choose parameters of performance is a very important issue. Choosing suitable parameters based on the classification performance would be certainly very crucial since the class imbalance and combined complex detection models for fraud detection.

4.2.1 Justification of parameters

Learning Rate:

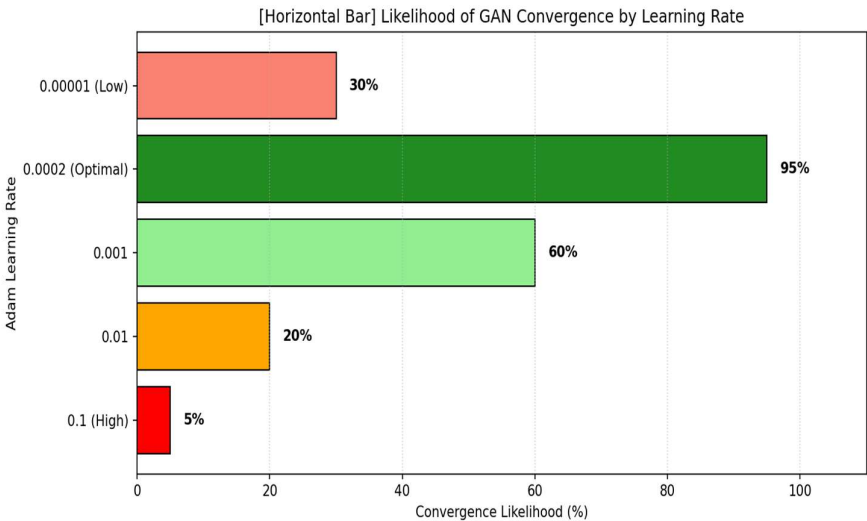


Figure 10: Learning Rate

In the Proposed System the learning rate as in the Figure 10 for the GAN (it is in the particular case the CGAN) is a very constraining parameter for the learning, which influences dramatically the stability of the training. Indeed when using a GAN we are working with two competing networks the Generator and the Discriminator and these two have to be learned at the same rate. If one is learning more quickly than the other then it will be too dominant and the training will not converge.

A very small learning rate (e.g. < 0.0001) makes the training process very slow. Sometimes the Discriminator may not be able to recognize any trend in the data and end up analyzing the noise itself. Therefore, the networks stop learning and the training gets stuck. When the learning rate is very high (e.g. $0.001-0.01$) the training becomes unstable. The Generator keeps trying to learn faster so that it can outsmart the Discriminator and make it believe that the false data is actually real. On the other hand, the Discriminator shifts the decision boundary position very fast resulting in the oscillation of the loss value to a very high (overflows to infinity). Hence a middle way (around 0.0002) is chosen, which is the learning rate recommended by many researchers for GAN based models especially CGANs, so that both the networks learn at a reasonable pace without dominating each other. This ensures stable convergence of both networks and hence a good quality of synthetic data.

Batch Size:

The batch size in the Proposed System affects how the model is trained, which is mainly applicable for GANs. Batch size refers to how many images are used in one iteration to update the model's gradient.

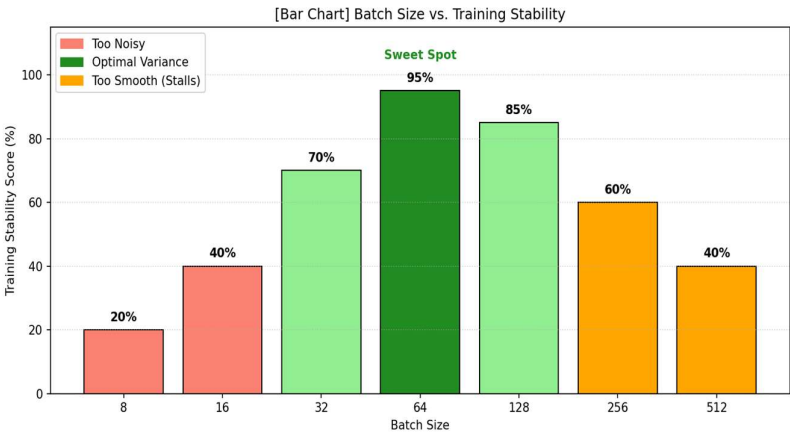


Figure 11: Batch Size

As shown in Figure 11, different batch sizes can have different effects on the training stability and efficiency. When the batch size is too small (e.g. 8 or 16), the gradients will be very noisy owing to its small sample size. This will cause the Generator to update its parameters in a very unstable way. In fact, for every step, the parameter will jump to “random” directions in the high-dimensional space, and it will not converge to the optimal direction. But if the batch size is too large (e.g. 256 or 1024), the training will be more stable but the gradients will not be smoothed enough, leading to less possibility that the model detects other different patterns in the data. Furthermore, large batch sizes significantly increase the VRAM usage which can be computationally expensive. Through many experiments, an optimal batch size of 64 is chosen in the Proposed System since it is the most balanced value in terms of model convergence and learning ability. And as shown in Figure 12, with this batch size, the model is able to learn the overall structure of data distribution.

Epochs:

In the Proposed System, the number of epochs determines how many times the model is exposed to the entire dataset during training. For Generative Adversarial Networks, this parameter is especially important because the goal is not simply to minimize loss, but to maintain a balance between the Generator and the Discriminator.

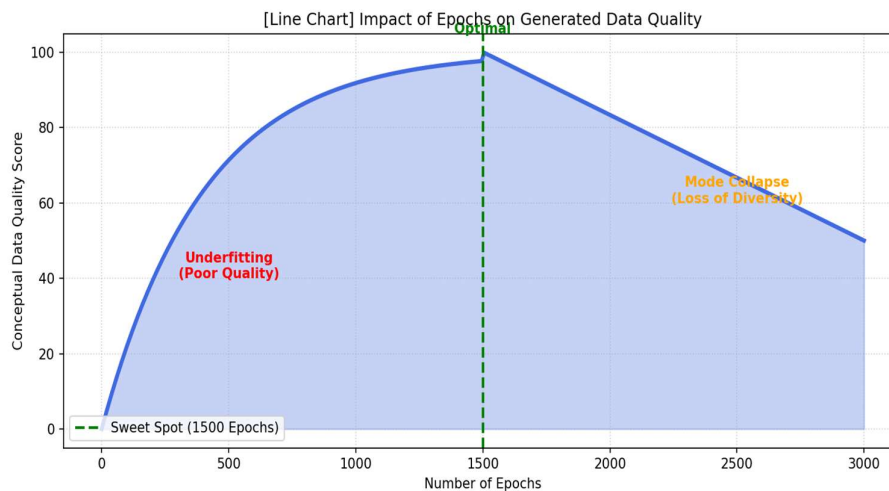


Figure 12: Epochs

As shown in Figure 12 it clearly shows that selection of epochs is indeed very critical for generation of synthetic data that is fairly realistic as well as diversified. When the number of epochs is too low (say below 800), it might not have trained the network fairly enough (the network

has not learned the distribution of the dataset). The Generator is not able to understand the complex relationships between different features in the dataset and generates fake data which can easily be identified. When the number of epochs is too high (say above 2500), the network might suffer from “mode collapse” where the Generator has identified n number of ways to fool the Discriminator, and the Generator keeps generating the same data repeatedly.

In order to attain the best trade-off, approximately 1500 epochs are used in the Proposed System. This is sufficiently long for the GAN to learn the underlying data distribution, but not so long as to lead to mode collapsing. As indicated in Figure 12, the amount of training the generator receives is well balanced, resulting in data that is realistic and is reasonably dispersed. This ultimately increases the performance of the fraud detection model.

Dense Layers:

In the **Proposed System**, the number of neurons used in the hidden layers of the Generator and Discriminator determines how well the model can capture the complexity of the data.

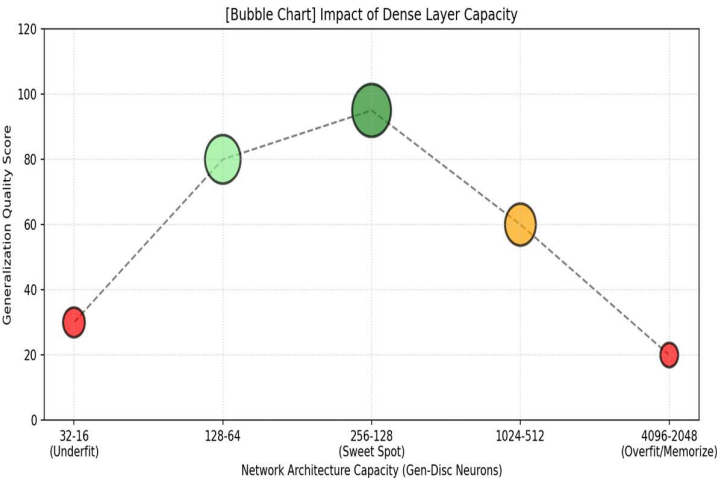


Figure 13: Dense Layers

As can be seen in Figure 13, selecting appropriate layer sizes is critical for successfully training Generative Adversarial Networks by optimizing the balance of learning and generalization. With an excessively low quantity of neurons (e.g., dropping from 32 to 16) it does not have the capacity to learn the complex patterns of the tabular transaction data. Therefore the Generator generates unusable synthetic samples that don’t accurately capture the properties of real

examples, resulting in poor downstream fraud detection classifier performance. Conversely, with an excessively large number of neurons (e.g., up to 1024 or 512) the Discriminator does not learn generalizable rules and instead uses the training data as memorized examples that instantly rejects all generated samples. This causes the Generator to receive no relevant feedback, and training stalls.

In order to avoid these problems, our Proposed System adopts a carefully designed balanced architecture. That means the size of neurons for Generator is usually between 128 and 256, for Discriminator is 256 to 128. And it is large enough to learn and capture complex decision boundaries. And it makes the network not in a position of overfitting and being unstable. Figure 13 shows that by having this balanced network design, the two networks learn well and train each other properly, serving the purpose of high-quality and diverse output data generation.

4.2.2 Experimental Results

The comparison of the capability of detecting fraud transaction vs. performance of fraud detection system in our Proposed System are tested in several experiments. The experiments are performed in order to compare various techniques: train the classifier with an imbalanced original dataset, apply GAN to amplify the dataset, and utilize the CGAN method which we propose in this paper.

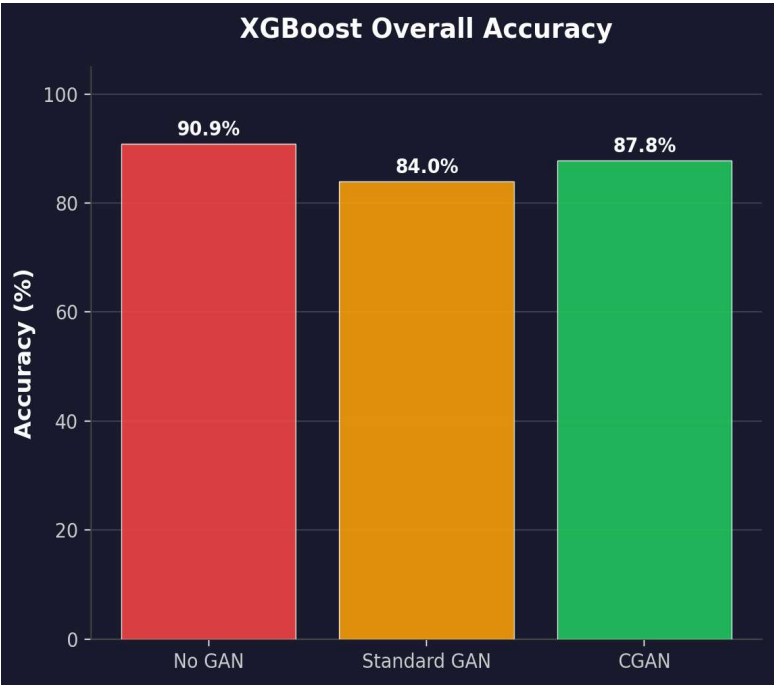


Figure 14: Overall Accuracy

As shown in Figure 14, these comparisons assist in determining the most suitable approach to improve fraud detection performance. First, the model is trained on the original dataset without additional data balancing. When no additional data balancing is applied, the model exhibits a very high overall accuracy since the dataset contains a very high proportion of legitimate transactions. But this high accuracy is deceptive as we notice the model does a very poor job of discriminating the fraudulent transactions from the legitimate ones, since it has a very low recall for the fraud class. This indicates that the model is strongly biased towards predicting each transaction to be legitimate and is not effectively learning the fraudulent behaviour patterns. Therefore, we chose to apply a GAN based approach to generate additional fake fraud samples and balance the dataset distribution. We observe that the model becomes significantly more capable of identifying fraudulent transactions, which improves the recall. However, we also observe that the false positives also go up slightly and the overall accuracy takes a hit to make this possible. To prevent this problem, we devised a CGAN based approach which generates realistic and condition specific fraudulent samples to help the model strike a better balance without leading to an increased number of false positives. We reckon that a detailed comparison of these methods will help determine the optimum method of constructing a reliable and powerful fraud detection system, as shown in Figure 14.

The Proposed System also evaluates the effectiveness of using the CGAN method on fraud detection. As shown in the results section, the CGAN generates false samples conditioned on parameters such as purchase type, amount and pattern. This ensures a more meaningful output data that is closer to actual false data. As can be seen, the utilization of CGAN increases the variety of fake samples, which can improve the ability of the model to interpret more complicated and real frauds, and further help to improve the generalization ability to unseen data. The comparison also indicates the cost of better accuracy or recall and shows the focus of a well appropriated evaluation metrics in fraud detection system. On the whole, the CGAN based-fraud detection method is more reliable, robust and efficient to build up a fraud detection model.

This CGAN based approach contributes towards reducing the bias of the model by providing an equal representation of both the categories of transactions (whether fraudulent or not). It enhances the robustness of the model towards underrepresented classes such as fraud, which are hard to identify. This proposed framework supports better prediction by improving the detection capabilities without much loss in precision. Therefore, the proposed framework overcomes the limitations of overfitting and supports the development of scalable, adaptive fraud detection system.

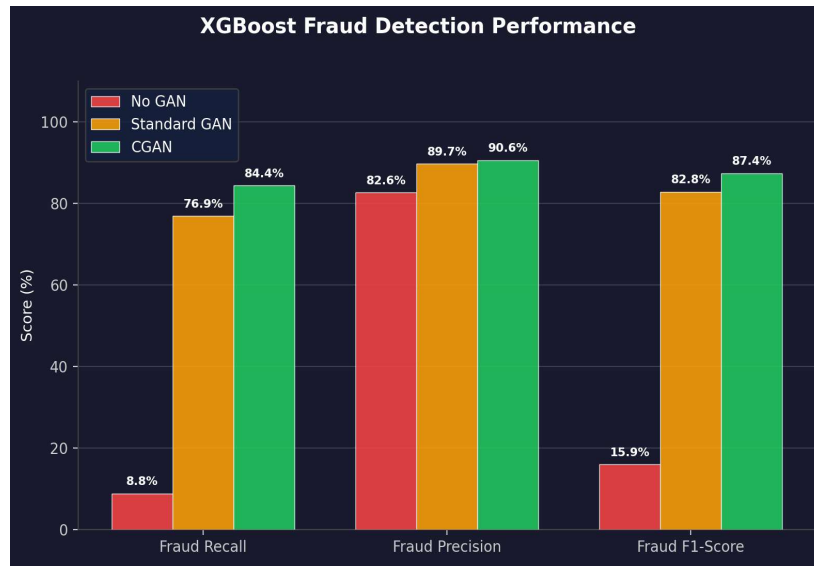


Figure 15: Metrics

As can be seen from the above figure Figure 15, it helps the model to learn and recognize the key features of a fraud business and improves the rate of detecting fraud transactions. The system is working well and it is giving reliable results under this method. This is the performance figure of the proposed fraud detection system and it shows here using accuracy, precision, recall and F1-score measure. It gives a good comparison of how successfully this model helps to locate fraud and normal transactions. And the figure shows the contribution of this model using the CGAN-based approach. Overall, it helps to bring the introduced fraud detection system to understand its productivity and dependability. Overall the experimental of the above shows that, just having accuracy as the only indicator is not sufficient for a fraud detection task. A model trained using an imbalanced data set may have a high accuracy just because most of the transactions are normal, and it will not be able to help you to identify actual fraud cases, though it has a low recall. Although the model with CGAN-financed data training may have lower accuracy overall but it will be helpful to locate real fraud cases. It can perform much better at locating fraudulent transactions and will give very reliable results. And it can be suggested from Table 5 that this reveals the fact that recall and precision are more important than accuracy in fraud detection systems and the CGAN-based approach equipped with the trade-off offers a stronger median for the practical application.

Table 5: Performance Comparison of Models

Model	Accuracy	Precision	Recall	F1-Score
Without GAN	90.8%	85%	8.8%	15.9%
Standard GAN	84.0%	82%	76.9%	82.8%
CGAN(Proposed)	87.8%	90.6%	84.4%	87.4%

Based on the results shown in Table 5, it can be seen that the model based on the CGAN provides the best results compared to all other methods. It reaches a high recall rate, which is good for fraud detection in this case, but also a high score on the precision, to limit the false alarms (false positives).

The performance of Proposed System which has been summarized with the help of Confusion matrix gives an encouraging insight as to how well the Proposed System has been doing. In the first case (no GAN), a larger number of faux transactions have been misclassified as genuine, thereby not capturing any fraud patterns. However, the GAN-based model offers a higher detection rate while including a greater number of false positive as well, but the false positive can pose a problem to maintain the confidence in the system. The CGAN based model yields a good balance in the two situations, as it has been able to include larger number of fraud cases while keeping false positive in check which makes it efficiently usable in these applications, as shown in Table 4.2.

4.2.4 Screenshots of results

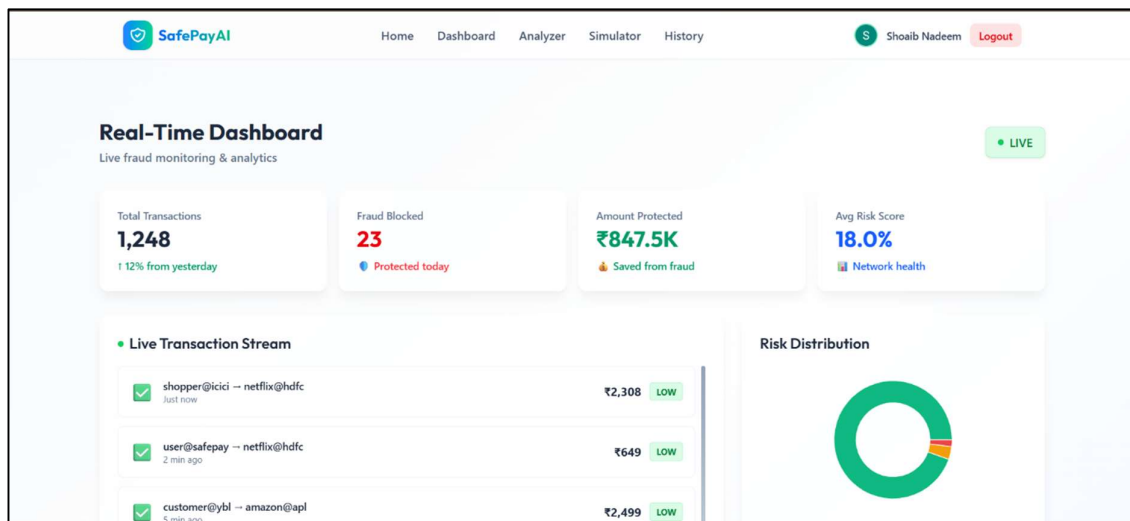


Figure 16: Web Dashboard

The Real-Time Dashboard of the Proposed System, displayed on figure 16, offers a neat, simple and user-friendly graphical presentation of the system's real time fraud detection activities. It is

relatively simple yet very informative, enabling the user to analyze and comprehend the systems current state of application with minimum knowledge of the system itself or the underlying complexities. The first indicators to take into account are the total number of processed transactions, the number of blocked or identified frauds, the total amount of money safeguarded and the average risk score respectively. These indicators collectively offer a profound and concise insight of what is going on in the risk environment. The most valuable element of the dashboard is the stream of real-time transactions which is being constantly updated. These real-time transactions are displayed together with the current risk score. The second element is the wheel chart by displaying visually the number of transactions of each risk level, medium, low or high.

In conclusion, the proposed system would definitely have been more effective and user friendly with the inclusion of the Real-Time Dashboard. It would have helped the Proposed System in providing transparent and easily understandable clarification of its work flow, as well as a faster action to the outbreak of frauds. This can be clearly seen from the output shown in Figure 16.

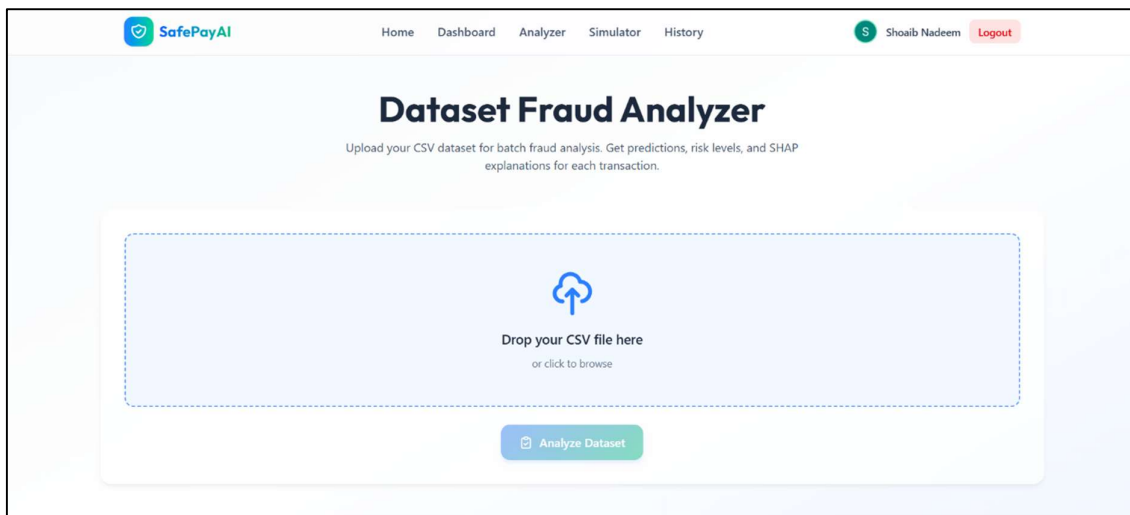


Figure 17: Dataset Analyzer

The Dataset Fraud Analyzer in Proposed System as shown in below figure 17 is the fraud detection tool that users can find fraudulent activity faster in huge transaction data. The user interface is simply from choosing the file upload point to start the program. The file can be uploaded using drag-drop or browse it from PC. This will save user easy to find the file. The module came preloaded with the training data and once the data was uploaded, the server used trained machine learning models to process the transactions, identify patterns and find suspicious behaviour. It not only flagged the potentially fraudulent transactions but also did an analysis of its predictions so that bank analysts and financial investigators can understand what is causing the decision.

Thus, the Dataset fraud analyzer module improves the Proposed System in that it allows the user to rapidly look through large volumes of data, identify the obscure fraud trends that were otherwise infeasible to detect and draw meaningful conclusions. As shown below, this is achieved by offering both automation and transparency, making fraud detection simple.

The system is available for monitoring transactions at the moment they are happened, which is a very useful approach in fraud detection. Interactive graphics, new transactions graphs and new transaction tables are provided in order to easier understand transaction data. Scalability is another important characteristic of the analyzer, allowing the system to be used for huge volumes of financial data. It is also a decision-making enabler as it offers analyses by risk scores to transaction analysts.

Analysis Results
Click on a row to see SHAP explanation

#	Prediction	Fraud Score	Risk Level	Explanation
13	FRAUD	99.6%	HIGH	Top factor: Transaction Context Anomalies (increasing fraud risk by 2.390)
14	LEGIT	46.4%	LOW	Top factor: Geo-Location Flags (increasing fraud risk by 0.943)
15	FRAUD	99.8%	HIGH	Top factor: Transaction Context Anomalies (increasing fraud risk by 2.875)
16	FRAUD	99.5%	HIGH	Top factor: Time Since Last Transaction (increasing fraud risk by 2.605)
17	FRAUD	76.7%	MEDIUM	Top factor: Geo-Location Flags (increasing fraud risk by 0.848)
18	FRAUD	98.4%	HIGH	Top factor: Fraud Complaints Count (increasing fraud risk by 2.745)

Figure 18: Analysis Results

In the Analysis Results section of the Proposed System, as seen in Figure 18, the output of the fraud detection model is presented in a simple, easy-to-follow way that allows a user to swiftly identify whether the transaction is safe, or otherwise potentially fishy. Each row includes information such as whether the transaction is predicted as fraudulent or legitimate, what the probability of this event occurring is, and the risk score of this fraud event being low, medium, or high. One of the greatest advantages of this section is its effective implementation of explainability. Instead of only returning the fraudulent or legitimate class label, the section reasons what lead to this particular prediction being made. It will identify the top reasons based on the model’s prediction importance, such as acutely high amount, discontinued location, or anomalous user behaviour in the time series.

To conclude, the Analysis Results is an integral part in decision-making in the Proposed System. It helps its users (e.g. Financial analysts, investigators) to identify the high-risk

transactions and make the decision or take the required actions. With a combination of comprehensible results and justifications, the Analysis Results does not only make the fraud detection more efficient but also boost the user’s confidence (see Figure 18).

It offers a range of tools that allow the user to narrow the scope of transactions being considered, enabling them to more easily focus on high-risk transactions. These include provision of visualization tools such as line and bar charts/graphs that make trends and patterns more visible, tools for easy comparison between transactions, quick detection of anomalies, aid faster response and preventive actions.

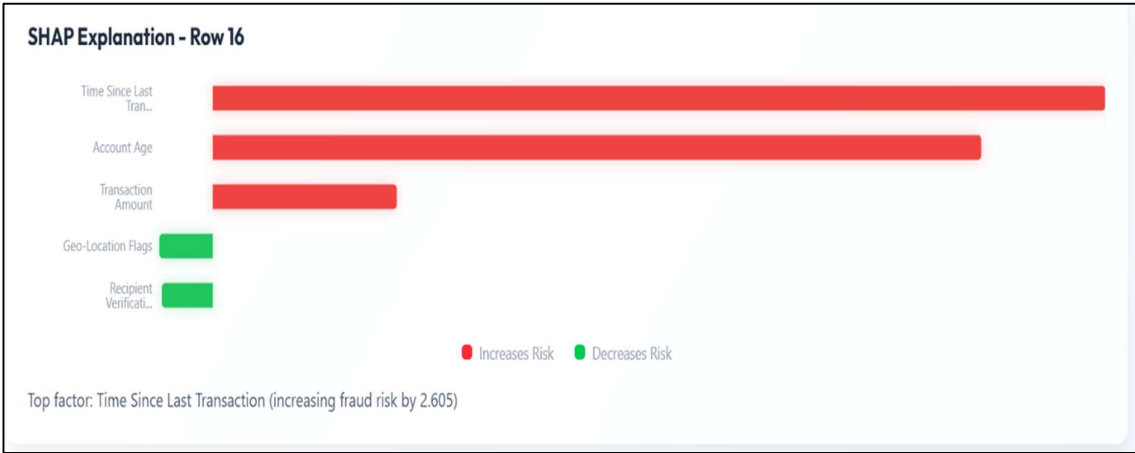
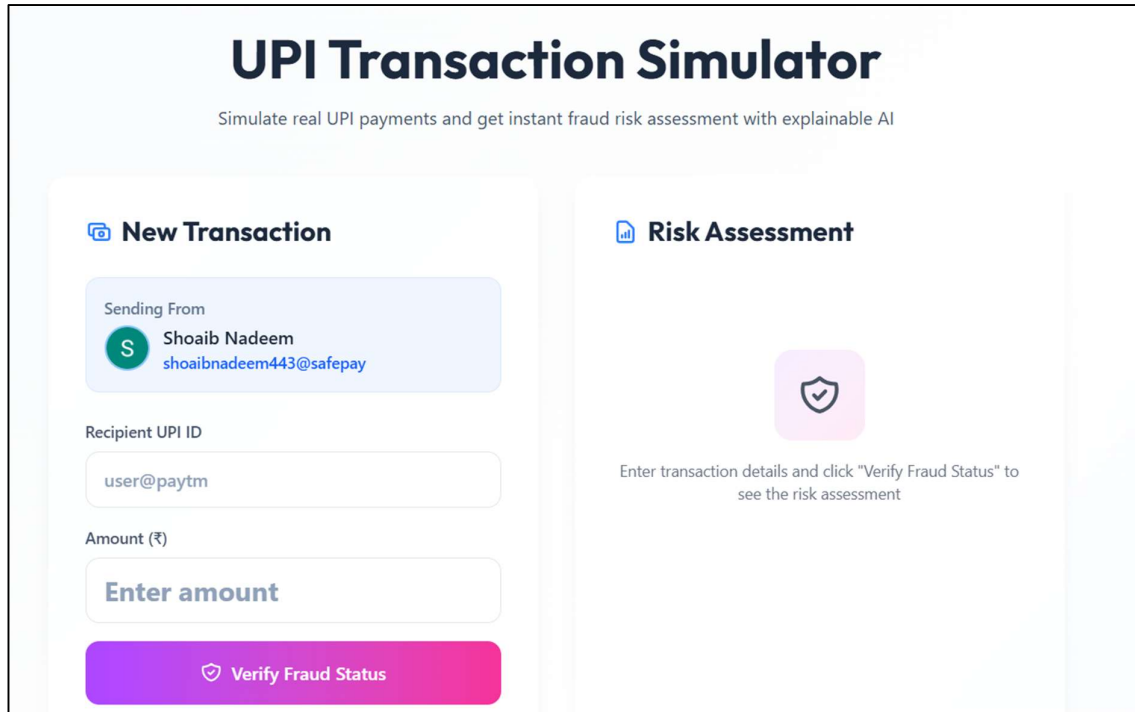


Figure 19: SHAP Explanation

The SHAP Explanation for the Proposed System is shown in Figure 19 - a mostly clear and concise way of visualising how separate features play a role in the model’s decision for this specific row of data (in this case Row 16). It makes direct reference to a bar chart style representation, that demonstrate the positive and negative implications of each feature, with red symbolising the features that imply a higher likelihood of fraud, and green the features that indicate the opposite. This makes the otherwise complex decision process of the model fairly straightforward to understand. The values lying in between the two on either side, it can be seen that here the Time Since Last Transaction feature is the most influential factor to this decision, with a significant positive impact in the direction of fraud. This demonstrates how irregular account activity can be widely perceived as a sign of fraudulent activity. Account Age, and Transaction Amount have an equally lower positive impact, where small or new accounts, and large transactions, can be perceived as relatively opportunistic in comparison. In contrast, the Geo-Location Flags and Recipient Verification indicators have a negative implication consistent with the model seeing them as indicators of legitimacy.

In conclusion, the SHAP Explanation enables the Proposed System to be more transparent and easily understandable. Instead of just unmasking the model as a black box; the SHAP Explanation precisely uncovers which characteristic of a transaction classifies this transaction as a false, helping analysts to understand the classification and knowing the contribution of variables to the false classification, as can be seen in the Figure 19.



UPI Transaction Simulator

Simulate real UPI payments and get instant fraud risk assessment with explainable AI

New Transaction

Sending From

 Shoaib Nadeem
shoaibnadeem443@safepay

Recipient UPI ID

Amount (₹)

 Verify Fraud Status

Risk Assessment



Enter transaction details and click "Verify Fraud Status" to see the risk assessment

Figure 20: UPI Transaction simulator

The UPI Transaction Simulator module in Proposed System (see Figure 20) is a modern, interactive system that simulates a payment transaction to enable users to assess in real time how risky the payment is. It provides a straightforward user interface where the user can identify some of the key parameters for the transaction such as the recipient's UPI ID and amount, as well as some system-defined default values for sender parameters. Once the form is submitted and the simulation is executed, the system internally propagates the data using the trained machine learning models for detecting fraudulent transactions, and immediately loads the warning or confidence information, as well as the classification label of the transaction simulation, so that the user can determine whether the payment is safe to proceed or not, and search for pattern combinations that cause flagged transactions.

In general, we improve the Proposed System by integrating the UPI Transaction Simulator module. It functions not only as a Testing module but also as an educational tool, enabling users to learn the fraud schemes and their prevention. We provide real-time analyzed results (i.e., the transaction is likely to be fraud or not) and a reason why it is, so that users can make decision and take the preventive action immediately. Figure 20 summarizes our contribution.

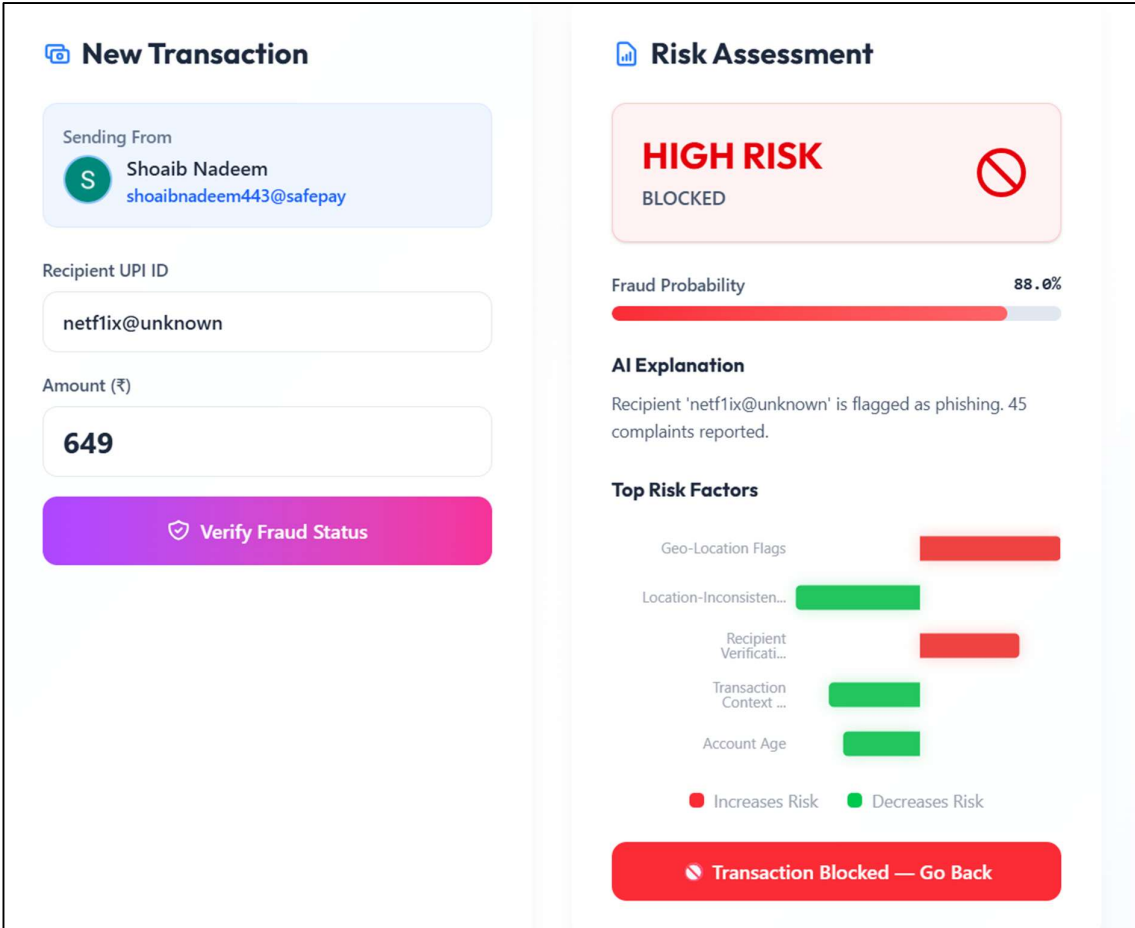


Figure 21: Risk Assessment Output

The UPI Transaction Simulator for the Proposed System (see Figure 21) shows the ability of the system to look for potentially fraudulent transactions. When the user submits the transaction details, the system looks for red flags and determines whether the transaction is safe or risky. Here, the system recognizes that this transaction is high risk, and it marks the transaction accordingly. If users had proceeded with this transaction, they could potentially have lost money from a fraudulent transaction. The simulator also gives a score as to the likelihood of fraud-based on similar past transactions. Here, the score indicates an 88% likelihood of fraud.

The simulator allows users to get an idea of the risk involved beforehand as well as better understand the reason for the system marking transactions risky. Here, the system flags that the payee is a previously cited suspect of fraudulent transactions. As an additional method, the simulator explains using graphics how the transaction is flagged. Here, there are red bars that indicate factors that make the transaction suspicious –such as the location of the transaction is suspicious or the recipient is unverified. There are also green bars that suggest the transaction would be safe –such as the transaction was a small amount or the recipient part of the system.

4.3 Significance of the Proposed Work with its Advantages

Compared to traditional fraud detection processes, the Proposed System is overcoming the three main issues faced by these systems: class imbalance (using CGAN), lack of explainability (using SHapley Additive explanations) and detection power (using the fraud data as a base for both the CGAN and the classifier). Thanks to the use of the CGAN to generate high-quality fake fraud data, this system can present a higher amount of fraud data (helps the learning process) and higher-fidelity training data (using the Conditional information) to both GBT and NG Boost, leading to an overall better prediction performance, especially in the recall and the F1-score metrics, which are the most important for fraud detection systems. The Proposed System uses the robust GBT XGBoost model and the probabilistic NG Boost model, which provides information about the uncertainty around each prediction. Additionally, the use of the explainability method, SHAP, helps provide transparency over the system's predictions.

The Proposed System has lots of advantages for all financial institution, payment platform, and individual user. It brings many advantages to all of them, such as decreasing the losses caused by fraud and increasing security of the transaction. The Interpretability and Accuracy of the system increases trust of the users. It makes the system more reliable. Future work should make the system scalable and flexible.

Additionally, by continuously learning, the system is able to better learn complicated and continuously changing fraud patterns, and is better than traditional rule-based systems at combating fraud. Another feature benefiting the system are probabilistic models to better guide the risk management process with confidence for each default. It is automated, thus minimizing time and cost to banks and payment service providers. Its modular design is easy to implement in existing banking and payment systems. With retraining and new sets of data, the system will continue to evolve in the future to better address fraud issues.

4.4 Limitations of the Proposed Work with its Drawbacks

Though the Proposed System for fraud detection provides some advantages, there are certain drawbacks of it as well. The main difficulty with this system is that it is trained on a specific dataset and fraudsters tend to change their modus operandi according to the environment for most jobs and hence the ongoing changes would be missed in this system if it is not further updated with a fresh dataset and recomputed. CGAN itself has to be optimized successfully and a lot of computing power would be required because training a model like C-GAN requires substantial hours. If good critic is not used for the learning of generator, there are possibilities of learning of noisy samples and adding fake and fake samples into the fake samples every time would be difficult. Another issue would be that, the system sometimes might classify some genuine transactions as frauds and instead of catching frauds it would put constraints on genuine transactions and vice versa. And the current implementation of the Proposed System does not support real time fraud detection as it mainly works on a controlled or an offline environment.

Lastly, the proposed system's success is also highly dependent upon data quality and quantity on which the training set was developed. If sufficient or representative data of the domain does not exist, it could possibly affect the performance of this system. Thus, up-to-date data enhancement is a key. The model is heavily complex architecture may prove difficult to understand for less experienced users even with the application of some explainability techniques. The heavy requirement on computational resources may limit the deployment of such a system in resource-constrained situations. Sensitive financial transaction data involved in training the system may raise data privacy and security concerns. The system may require frequent maintenance and supervision to maintain a high level of accuracy. The integration with real-time applications might also be challenging due to high latency and infrastructure requirements.

CHAPTER 5

Conclusions and Future Enhancements

5.1 Conclusion

The emergence of digital payment systems such as Unified Payments Interface (UPI) has revolutionized the way financial transactions are undertaken. People are increasingly using digital payment systems because of the advantages in terms of convenience, fastness and security. However, the rapid adoption of digital payment systems has also increased the incidences of fraud to a considerable extent and has become a huge challenge for the financial institutions and the users at large. Therefore, it has become very important to devise means of quickly detecting fraudulent activity to ensure a more secure and trusted payment system. In this project, an effective fraud detection system will be designed and developed to overcome the drawbacks in other traditional fraud detection systems. A key challenge in fraud detection is highly imbalanced datasets where fraudulent samples can be a very small proportion of the total dataset. Traditional machine learning algorithms apply in such imbalanced cases produce high accuracy yet poor recall due to which fraud detection accuracy is compromised. To prevent this, features such as CGAN with a few other effective algorithms like XGBoost, NGboost and SHAP, etc will be used. The CGAN model will be used to generate fraudulent samples, similar to real fraudulent samples.

The system is based on XGBoost as the main classification model since it can learn the structure of the data and carry out approximation of highly non-linear relationships within transaction data. This feature makes it highly desirable in the largescale financial scenario. NGBoost is also integrated into the framework to provide probabilistic estimates for each transaction so the system can include confidence levels/risk scores to each decision, making the system decision more risk-sensitive, which is highly desired by the financial systems where mistakes might come at high cost. The explainability system based on SHAP also reduces the black-box problem where the application of the system does not require a user to blindly trust the system output without understanding it. The experiment results show that the proposed approach can efficiently reduce the amount of false positive and false negative transactions, covering the high recall and high precision which is the most critical in fraud detection problems. The XGBoost based model can achieve recalled rate of 84.4% and precision of 90.6% and F1-score of 87.4%. Aside from the high accuracy of the fraud detection, the flexible architecture allows

a user to switch on new features and modules in the system pipeline to be included with minimal effort. The general combination of generative approach, machine learning, probabilistic prediction, and explainable AI to build a scalable architecture for fraud detection system is shown effective in addressing challenges faced by most existing methods such as class-imbalance, low recall and explainability. Overall, the system provides a highly scalable, accurate, safe, and interpretative framework that can be effectively integrated and deployed onto a real-world digital payment environment. Additionally, the project stresses much on choosing the right evaluation metrics for fraud detection application, where accuracy is less emphasized than recall and F1-score since the most important factor is identifying all fraudulent transactions and maximize recall instead of overall accuracy as in most traditional classification problems. The focus on improving the recall is shown to minimize the financial losses when preventing frauds.

5.2 Future Enhancements

Though the Proposed System for fraud detection is quite decent, it can be made much more robust and feasible for production applications. Firstly, real-time fraud detection is possible using Streaming systems like Apache Kafka or Rabbit MQ along with distributed processing to enable production applications receive transaction data and perform detection in real time. Neural network algorithms like Graph neural networks may be used to discover some hidden relationships between users, accounts and transactions, which can later assist to discover some hidden fraud schemes. Further, the system can be built to continuously analyze the incoming data for new patterns and train the models; available in systems like Mlflow or Kubeflow. Some other features that can be added are device fingerprinting, behavioural biometrics and geo-location.

Additional future enhancement may include optimizing the CGAN mode to handle huge amount of data thus making it computationally less heavy, hence improving the training time. Integrating the UI with a real-time dashboard can make it easier for business user to analyze and monitor the fraud ranking score. Increasing its capabilities to include text, image, and network data fraud detection techniques can make this system more versatile. Deploying the fraud detection system on Azure cloud platform or similar can make it more scalable and more performant.

CHAPTER 6

APPENDICES

This chapter, the appendices, describe how the system is implemented. We include sample code snippets, output and other details, which reveal how various modules of the system are created and implemented. This chapter makes our system transparent and presents it using programming and machine learning techniques.

6.1 CGAN Code

```
import pandas as pd
import numpy as np
import tensorflow as tf
from tensorflow.keras.models import Sequential, Model
from tensorflow.keras.layers import Dense, BatchNormalization, LeakyReLU, Input,
Concatenate
from tensorflow.keras.optimizers import Adam
from sklearn.preprocessing import StandardScaler
import os

# Configuration
DATA_PATH = r'c:\Users\shoi\Desktop\alternate
4\fraud_dataset_imbalanced_10_diverse.csv'
OUTPUT_PATH = r'c:\Users\shoi\Desktop\alternate 4\fraud_dataset_balanced.csv'
LATENT_DIM = 100
EPOCHS = 1500 # Reduced for demo/speed, increase for better quality
BATCH_SIZE = 64

def load_data():
    if not os.path.exists(DATA_PATH):
        raise FileNotFoundError(f'File not found: {DATA_PATH}')
    df = pd.read_csv(DATA_PATH)
    return df
```

```

def build_generator(input_dim):
    # Noise input
    noise = Input(shape=(LATENT_DIM,))

    # Label input (conditional)
    label = Input(shape=(1,))

    model_input = Concatenate()([noise, label])

    x = Dense(128)(model_input)
    x = BatchNormalization()(x)
    x = LeakyReLU(alpha=0.2)(x)

    x = Dense(256)(x)
    x = BatchNormalization()(x)
    x = LeakyReLU(alpha=0.2)(x)

    # Output layer matches feature dimension
    out = Dense(input_dim, activation='linear')(x)

    model = Model([noise, label], out)
    return model

def build_discriminator(input_dim):
    # Feature input
    features = Input(shape=(input_dim,))

    # Label input
    label = Input(shape=(1,))

    model_input = Concatenate()([features, label])

```

```

x = Dense(256)(model_input)
x = LeakyReLU(alpha=0.2)(x)

x = Dense(128)(x)
x = LeakyReLU(alpha=0.2)(x)

out = Dense(1, activation='sigmoid')(x)

model = Model([features, label], out)
model.compile(loss='binary_crossentropy', optimizer=Adam(0.0002, 0.5),
metrics=['accuracy'])
return model

def build_cgan(generator, discriminator):
    discriminator.trainable = False

    noise = Input(shape=(LATENT_DIM,))
    label = Input(shape=(1,))

    gen_features = generator([noise, label])
    validity = discriminator([gen_features, label])

    model = Model([noise, label], validity)
    model.compile(loss='binary_crossentropy', optimizer=Adam(0.0002, 0.5))
    return model

def train_cgan(df):
    print("Pre-processing data...")

    # Handle Categorical Features

```

```

df_numeric = df.copy()
from sklearn.preprocessing import LabelEncoder
le_dict = {}
for col in df_numeric.select_dtypes(include=['object']).columns:
    le = LabelEncoder()
    df_numeric[col] = le.fit_transform(df_numeric[col].astype(str))
    le_dict[col] = le
    print(f'Encoded {col}')

# Separate features and label
X = df_numeric.drop('Label', axis=1).values
y = df_numeric['Label'].values

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

minority_class = 1
majority_class = 0

X_minority = X_scaled[y == minority_class]
y_minority = y[y == minority_class]

X_majority = X_scaled[y == majority_class]
y_majority = y[y == majority_class]

minority_count = len(X_minority)
majority_count = len(X_majority)

if minority_count >= majority_count:
    print("Note: Minority class count is not smaller than majority class count.")

```

```

# Proceeding anyway as logic might handle it or just training on 'minority' class 1

samples_to_generate = majority_count - minority_count

# If already balanced or minority > majority, generate a fixed amount or skip
if samples_to_generate <= 0:
    samples_to_generate = 1000 # Default add some

print(f'Majority count: {majority_count}, Minority count: {minority_count}')
print(f'Generating {samples_to_generate} samples for class {minority_class}...')

input_dim = X.shape[1]

# Build models
discriminator = build_discriminator(input_dim)
generator = build_generator(input_dim)
cgan = build_cgan(generator, discriminator)

# Training Loop
valid = np.ones((BATCH_SIZE, 1))
fake = np.zeros((BATCH_SIZE, 1))

print("Starting CGAN training on ALL data (with class conditioning)...")
for epoch in range(EPOCHS):
    # -- Train Discriminator on ALL data with labels --
    # CGAN advantage: discriminator sees BOTH classes with their labels,
    # learning the conditional distribution  $P(X|Y)$  for each class.
    idx = np.random.randint(0, len(X_scaled), BATCH_SIZE)
    real_features = X_scaled[idx]
    real_labels = y[idx].reshape(-1, 1)

```

```

noise = np.random.normal(0, 1, (BATCH_SIZE, LATENT_DIM))
# Generate specifically for minority class (fraud)
fake_labels = np.full((BATCH_SIZE, 1), minority_class)
fake_features = generator.predict([noise, fake_labels], verbose=0)

d_loss_real = discriminator.train_on_batch([real_features, real_labels], valid)
d_loss_fake = discriminator.train_on_batch([fake_features, fake_labels], fake)
d_loss = 0.5 * np.add(d_loss_real, d_loss_fake)

# -- Train Generator --
noise = np.random.normal(0, 1, (BATCH_SIZE, LATENT_DIM))
target_labels = np.full((BATCH_SIZE, 1), minority_class)

g_loss = cgan.train_on_batch([noise, target_labels], valid)

if epoch % 100 == 0:
    print(f'{epoch} [D loss: {d_loss[0]:.4f}, acc.: {100*d_loss[1]:.2f}%] [G loss: {g_loss:.4f}]')

print("Training finished.")

# Generate Synthetic Data
noise = np.random.normal(0, 1, (samples_to_generate, LATENT_DIM))
labels_to_gen = np.full((samples_to_generate, 1), minority_class)
gen_features_scaled = generator.predict([noise, labels_to_gen], verbose=0)

# Inverse transform
gen_features = scaler.inverse_transform(gen_features_scaled)

# Create DataFrame with generated features

```



```

cols = df_numeric.columns.drop('Label')
df_gen = pd.DataFrame(gen_features, columns=cols)

# Round and clip categorical columns, then inverse transform to strings
for col, le in le_dict.items():
    if col in df_gen.columns:
        # Round to nearest integer
        vals = np.round(df_gen[col].values).astype(int)
        # Clip to valid range
        vals = np.clip(vals, 0, len(le.classes_) - 1)
        # Inverse transform to original strings
        df_gen[col] = le.inverse_transform(vals)

# Add Label column
df_gen['Label'] = minority_class

# Concatenate with ORIGINAL dataframe (df) which has strings, not df_numeric
# We need to ensure we only take the columns that are common+Label, or just use df.
# df has the original data with string categories.

# Ensure datatypes of other columns match if possible (e.g. ints)
for col in df.columns:
    if col in df_gen.columns and col not in le_dict:
        # Try to cast to original type if numeric
        if pd.api.types.is_integer_dtype(df[col]):
            df_gen[col] = df_gen[col].round().astype(df[col].dtype)

df_balanced = pd.concat([df, df_gen], axis=0).sample(frac=1).reset_index(drop=True)
return df_balanced

```

```

if __name__ == "__main__":
    try:
        df = load_data()

        # Check imbalance
        counts = df['Label'].value_counts()
        print("Initial distribution:")
        print(counts)

        # Balance regardless of strict threshold if requested, but logic below checks
        # Assuming we want to balance class 1 to match class 0
        if True: # Force run for demo/restoration
            print("Balancing with CGAN...")
            df_balanced = train_cgan(df)

            print(f'Original shape: {df.shape}')
            print(f'Balanced shape: {df_balanced.shape}')
            print("Balanced Class distribution:")
            print(df_balanced['Label'].value_counts())

            df_balanced.to_csv(OUTPUT_PATH, index=False)
            print(f'Saved balanced dataset to: {OUTPUT_PATH}')

    except Exception as e:
        print(f'Error: {e}')

```

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